

**UNIVERSITY OF FINANCE – MARKETING**

**FACULTY OF COMMERCE AND TOURISM**

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**GROUP 5**

**A HYBRID APPROACH TO LAPTOP SELECTION  
FOR OFFICE WORKERS IN VIETNAM**

**GROUP ASSIGNMENT REPORT**

**Ho Chi Minh City, July 2026**

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Subject: Logistics Management

**GROUP ASSIGNMENT REPORT**

Lecturer: Assoc. Prof. Nong Thi Nhu Mai

**Ho Chi Minh City, July 2026**

# GROUP MEMBERS

| Member               | Student ID |
|----------------------|------------|
| Nguyen Vu Quoc Trung | 2321002610 |
| Luong Gia Lap        | 2321002363 |
| Vu Manh Hieu         | 2321002325 |
| Tran Vo Khanh Hung   | 2321002338 |
| Nguyen Thuy Duong    | 2321002282 |

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Representative of the group

Nguyen Vu Quoc Trung

## Abstract

Laptop-selection studies consistently rank price, processing capability, and battery life among the criteria that matter most to buyers, yet the priorities they report are drawn from products, populations, and infrastructure conditions that do not automatically hold once the buyer, the catalogue, and the country all change. This study addresses that gap by developing and applying a three-stage decision model for office workers in Vietnam that combines the Analytic Hierarchy Process (AHP) with the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS). Candidate criteria were first synthesised from prior studies and refined through semi-structured interviews with five experts positioned across the procurement process, then screened through a survey of 84 respondents, retaining seven criteria: price, processor, memory, storage, brand, weight, and battery life. AHP weights were subsequently derived from the representative expert's pairwise comparisons and verified for consistency ( $CR = 0.048$ ); price (0.349) and processor capability (0.277) jointly account for 62.6% of the total priority, whereas battery life, the most highly rated criterion in the screening survey, receives the smallest weight, its status shifting from a valued attribute to a threshold already satisfied once mains power can be assumed. TOPSIS then ranked twelve laptop models drawn from seven professional review outlets using market data; the Lenovo ThinkPad T14 Gen 6 (AMD) ranked first ( $C_i = 0.6152$ ) without dominating any single criterion outright, instead combining a below-median price with a leading brand-share position into the most balanced profile in the set, while the cheapest and the most expensive candidates ranked seventh and last. By separating which attributes are locally relevant from how much each one should weigh, the model turns an implicit purchase decision into a reproducible and auditable procedure that procurement teams can update as catalogues change.

**Keywords:** Multi-Criteria Decision-Making; AHP; TOPSIS; laptop selection; office workers; Vietnam.

# Contents

|          |                                                                 |           |
|----------|-----------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                             | <b>7</b>  |
| 1.1      | Background and Motivation . . . . .                             | 7         |
| 1.2      | Problem Statement . . . . .                                     | 8         |
| 1.3      | Research Gap and Contributions . . . . .                        | 8         |
| 1.4      | Objectives and Research Questions . . . . .                     | 9         |
| 1.5      | Scope and Structure of the Report . . . . .                     | 9         |
| <b>2</b> | <b>Literature Review</b>                                        | <b>11</b> |
| 2.1      | Laptop Selection as a Multi-Criteria Decision Problem . . . . . | 11        |
| 2.2      | An Overview of MCDM Methods . . . . .                           | 11        |
| 2.3      | The Analytic Hierarchy Process . . . . .                        | 13        |
| 2.4      | The TOPSIS Method . . . . .                                     | 14        |
| 2.5      | Prior Studies on Laptop and Electronics Selection . . . . .     | 15        |
| 2.6      | Research Gap and Positioning . . . . .                          | 17        |
| <b>3</b> | <b>Methodology</b>                                              | <b>18</b> |
| 3.1      | The Proposed Research Model . . . . .                           | 18        |
| 3.2      | Stage 1 — Criteria Development . . . . .                        | 18        |
| 3.2.1    | Literature Synthesis . . . . .                                  | 18        |
| 3.2.2    | Expert Interviews . . . . .                                     | 19        |
| 3.2.3    | Survey-Based Screening . . . . .                                | 20        |
| 3.3      | Stage 2 — AHP for Criteria Weighting . . . . .                  | 21        |
| 3.4      | Stage 3 — Scenario-Based Application with TOPSIS . . . . .      | 23        |
| 3.4.1    | Decision Scenario . . . . .                                     | 23        |
| 3.4.2    | Alternative Identification . . . . .                            | 24        |
| 3.4.3    | Data Collection and Operationalisation . . . . .                | 25        |
| 3.4.4    | The TOPSIS Procedure . . . . .                                  | 26        |
| <b>4</b> | <b>Results</b>                                                  | <b>28</b> |
| 4.1      | Criteria Development Outcomes . . . . .                         | 28        |
| 4.1.1    | The Candidate Criteria Set . . . . .                            | 28        |

|          |                                                                      |           |
|----------|----------------------------------------------------------------------|-----------|
| 4.1.2    | Interview-Driven Refinement . . . . .                                | 28        |
| 4.2      | Criteria Screening Results . . . . .                                 | 30        |
| 4.3      | Criteria Weights and Consistency Check . . . . .                     | 31        |
| 4.4      | Decision Scenario and Alternatives . . . . .                         | 32        |
| 4.5      | TOPSIS Ranking Results . . . . .                                     | 33        |
| <b>5</b> | <b>Discussion</b>                                                    | <b>36</b> |
| 5.1      | Interpretation of the Criteria Weights . . . . .                     | 36        |
| 5.2      | Interpretation of the Ranking . . . . .                              | 38        |
| 5.3      | Practical and Managerial Implications . . . . .                      | 39        |
| 5.4      | Comparison with Prior Studies . . . . .                              | 39        |
| <b>6</b> | <b>Conclusion</b>                                                    | <b>42</b> |
| 6.1      | Summary of Findings . . . . .                                        | 42        |
| 6.2      | Contributions . . . . .                                              | 43        |
| 6.3      | Limitations . . . . .                                                | 43        |
| 6.4      | Future Work . . . . .                                                | 44        |
|          | <b>References</b>                                                    | <b>45</b> |
| <b>A</b> | <b>Theoretical Basis for the Candidate Criteria</b>                  | <b>48</b> |
| A.1      | Economic Factors and the Budget Constraint . . . . .                 | 48        |
| A.2      | Core Hardware Capability . . . . .                                   | 49        |
| A.3      | Portability and Durability . . . . .                                 | 49        |
| A.4      | Interface and Interaction Experience . . . . .                       | 50        |
| A.5      | Trust and Emotional Value . . . . .                                  | 50        |
| A.6      | Software Compatibility (Added After the Expert Interviews) . . . . . | 51        |
| A.7      | Summary . . . . .                                                    | 52        |
| <b>B</b> | <b>Expert Interview Protocol and Summary</b>                         | <b>53</b> |
| B.1      | Interview Protocol . . . . .                                         | 53        |
| B.2      | Expert Profiles . . . . .                                            | 54        |
| B.3      | Thematic Summary of Findings . . . . .                               | 54        |
| B.3.1    | Hardware Priorities and the GPU Question . . . . .                   | 54        |
| B.3.2    | Battery, Weight, and Design Trade-offs . . . . .                     | 55        |
| B.3.3    | Brand, After-Sales, and Divergent Views on Trust . . . . .           | 55        |
| B.3.4    | Locally Specific Factors . . . . .                                   | 56        |
| B.3.5    | Trade-off Scenario Responses . . . . .                               | 56        |
| <b>C</b> | <b>Likert Survey Questionnaire</b>                                   | <b>58</b> |

|          |                                                                          |           |
|----------|--------------------------------------------------------------------------|-----------|
| <b>D</b> | <b>AHP Pairwise Comparison Matrix and Step-by-Step Calculation</b>       | <b>59</b> |
| D.1      | Step 1 — Column Sums . . . . .                                           | 59        |
| D.2      | Step 2 — Weight Vector by Power Iteration (Eigenvector Method) . . . . . | 60        |
| D.3      | Step 3 — Consistency Check . . . . .                                     | 60        |
| D.4      | Robustness Check — Geometric-Mean Weights . . . . .                      | 61        |
| <b>E</b> | <b>Laptop Dataset, Benchmarks, and TOPSIS Step-by-Step Calculation</b>   | <b>62</b> |
| E.1      | Benchmark and Market-Share Conversion . . . . .                          | 62        |
| E.2      | TOPSIS Step-by-Step Calculation . . . . .                                | 62        |



# List of Figures

|     |                                                                                    |    |
|-----|------------------------------------------------------------------------------------|----|
| 2.1 | Families of MCDM methods. . . . .                                                  | 12 |
| 2.2 | The complete three-level AHP hierarchy. . . . .                                    | 13 |
| 3.1 | The proposed research model. . . . .                                               | 19 |
| 4.1 | Transition from the literature-derived criteria to the final retained set. . . . . | 29 |
| 4.2 | AHP weights of the seven retained criteria. . . . .                                | 31 |
| 4.3 | TOPSIS closeness coefficients of the twelve candidate laptops. . . . .             | 34 |

# List of Tables

|     |                                                                                   |    |
|-----|-----------------------------------------------------------------------------------|----|
| 2.1 | Prior studies on laptop and electronics selection using MCDM. . . . .             | 16 |
| 3.1 | The Saaty fundamental scale of pairwise comparison. . . . .                       | 22 |
| 3.2 | Random Index (RI) values by matrix order $n$ . . . . .                            | 23 |
| 3.3 | Review outlets used as sources for alternative identification. . . . .            | 24 |
| 3.4 | Operational definitions of the TOPSIS criteria. . . . .                           | 26 |
| 4.1 | The fourteen candidate evaluation criteria in six groups. . . . .                 | 29 |
| 4.2 | Descriptive statistics and screening decision for the candidate criteria. . . . . | 30 |
| 4.3 | Criteria weights obtained by AHP. . . . .                                         | 31 |
| 4.4 | Candidate identifiers assigned to the twelve laptop models. . . . .               | 32 |
| 4.5 | Candidate laptops and their raw attribute values. . . . .                         | 33 |
| 4.6 | TOPSIS ranking of the candidate laptops. . . . .                                  | 34 |
| 5.1 | The present priority structure set against prior studies. . . . .                 | 40 |
| A.1 | Candidate criteria and their supporting literature. . . . .                       | 52 |
| B.1 | Structure of the five-part interview protocol. . . . .                            | 53 |
| B.2 | Profiles of the five interviewed experts. . . . .                                 | 54 |
| C.1 | Items of the criteria-screening questionnaire. . . . .                            | 58 |
| D.1 | Pairwise comparison matrix of the seven retained criteria. . . . .                | 59 |
| D.2 | Column sums of the pairwise comparison matrix. . . . .                            | 59 |
| D.3 | Convergence of the power-iteration weight vector across six iterations. . . . .   | 60 |
| D.4 | Consistency check: $A\mathbf{w}$ and the ratio $(A\mathbf{w})_i/w_i$ . . . . .    | 60 |
| D.5 | Eigenvector weights versus geometric-mean weights. . . . .                        | 61 |
| E.1 | CPU-to-benchmark conversion key. . . . .                                          | 62 |
| E.2 | Brand-to-market-share conversion key. . . . .                                     | 63 |
| E.3 | Converted decision matrix submitted to TOPSIS. . . . .                            | 63 |
| E.4 | Euclidean norms of the criterion columns. . . . .                                 | 64 |
| E.5 | Normalised decision matrix $r_{ij}$ . . . . .                                     | 64 |

|     |                                                                     |    |
|-----|---------------------------------------------------------------------|----|
| E.6 | Weighted normalised decision matrix $v_{ij}$ . . . . .              | 65 |
| E.7 | Positive ( $A^+$ ) and negative ( $A^-$ ) ideal solutions. . . . .  | 65 |
| E.8 | Separation measures $S_i^+$ and $S_i^-$ for each candidate. . . . . | 65 |

# Chapter 1

## Introduction

### 1.1 Background and Motivation

Laptop selection is a recurrent decision for both individuals and organisations because the device combines a relatively short replacement cycle with a rapidly changing catalogue of brands, models, and configurations. Historical shipment evidence illustrates the scale of this decision, with Sönmez Çakır and Pekkaya (2020) reporting 161.6 million laptop units sold in 2017 and Gaur (2023) documenting the further expansion associated with remote work and study in 2020–2021. The relevance of these figures lies less in the temporary pandemic surge than in the persistent replacement problem created by a product life commonly estimated at approximately five years (Sönmez Çakır & Pekkaya, 2020).

Replacement does not reduce the decision to choosing the newest or fastest device. A prospective buyer must compare attributes measured on different scales, including price, processor performance, memory, storage, weight, battery capacity, and brand-related considerations. These attributes also pull against one another rather than varying independently. A lower price constrains hardware capability because a manufacturer can only fit so much component cost within a given margin, and a larger battery adds weight because the energy density of lithium-ion cells is bounded, so additional capacity has to come from a physically heavier pack. The proliferation of configurations therefore increases the information available to the buyer without necessarily making the choice more coherent. An explicit decision model is useful because it records how competing attributes are weighted and how each alternative performs against the same criteria.

A problem of this form falls within Multi-Criteria Decision-Making (MCDM). The systematic review by Basílio et al. (2022) documents the application of MCDM across a wide range of domains, with the Analytic Hierarchy Process (AHP) and the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) among the most frequently applied methods. Their prevalence is also visible in the bibliometric analysis of Zyoud and Fuchs-Hanusch (2017). AHP and TOPSIS perform different analytical roles. AHP derives a priority vector from pair-

wise judgements, whereas TOPSIS ranks alternatives by their relative distance from positive and negative ideal solutions. Used together, they separate the elicitation of preferences from the evaluation of products.

## **1.2 Problem Statement**

This study examines the decision faced by an office worker in Vietnam who must select a laptop for work. The study does not impose a universal price band, because budgets vary across individual and organisational purchases. Price is instead treated as a cost criterion within the decision model; in an applied procurement setting, a buyer may additionally use a budget ceiling as an eligibility rule before ranking the remaining alternatives. This distinction avoids assuming that one expenditure range represents every office buyer while preserving the role of affordability in the trade-offs that follow.

The office-worker context also affects how criteria are interpreted. Prior studies identify recurring attributes such as price, processing capability, brand, battery life, and portability, but their priorities are derived from different products, markets, and user populations (Liao & Chuang, 2022; Sönmez Çakır & Pekkaya, 2020; Weng Siew Lam et al., 2023). Those priorities cannot be assumed to transfer unchanged to Vietnamese working conditions, where local factors such as the ready availability of mains power in offices or the prevalence of motorbike commuting may change how much a given attribute contributes to the decision. The criteria set must first be examined against local experience and then screened by the intended user population before its relative weights are elicited.

Previous research demonstrates that hybrid MCDM models can rank laptops and related electronic devices. In particular, Gaur (2023) combine AHP and the Best–Worst Method for weighting with TOPSIS for laptop ranking, while Weng Siew Lam et al. (2023) apply an integrated AHP–TOPSIS model to mobile phones. The unresolved issue is therefore not whether AHP and TOPSIS can be combined. It is how a laptop-ranking model can maintain a traceable connection between literature-derived criteria, local expert interpretation, wider user screening, preference elicitation, and a date-stamped set of product alternatives.

## **1.3 Research Gap and Contributions**

The literature reveals a separation between three forms of evidence. Laptop-selection studies identify criteria and sometimes rank alternatives, but commonly begin from a predetermined criteria set. Consumer studies provide evidence about attributes that influence purchase, yet do not necessarily translate those attributes into an auditable ranking model. Context-specific qualitative evidence can reveal local constraints, but expert observations alone do not establish that the wider target population assigns the same importance to each criterion. Few studies join

these forms of evidence in a staged process that distinguishes criterion identification, population screening, preference weighting, and alternative ranking.

This study addresses that design gap through two contributions. First, it develops a context-calibrated criteria set for office workers in Vietnam by combining the international literature, interviews with five experts occupying different positions in laptop procurement, and a screening survey of 84 respondents. This sequence makes agreements and divergences between published evidence, professional judgement, and user ratings visible. Second, it implements a reproducible AHP–TOPSIS application in which one representative expert supplies the pairwise AHP judgements and TOPSIS ranks twelve internationally recommended laptop models using price and configuration data recorded from Amazon.com on 12 July 2026. The resulting ranking is an evaluation of an international candidate snapshot for a Vietnamese office-worker decision context; it is not presented as a census of the Vietnamese retail market.

## 1.4 Objectives and Research Questions

The study has three objectives. The first is to establish criteria appropriate for evaluating office laptops in the Vietnamese context. The second is to estimate the relative importance of the retained criteria through AHP. The third is to apply those weights in TOPSIS to rank a defined set of laptop alternatives. These objectives are expressed through the following research questions:

1. **RQ1:** Which criteria are appropriate for evaluating laptops for office workers in Vietnam?
2. **RQ2:** What weights do these criteria carry when determined through AHP?
3. **RQ3:** Under a simulated purchase scenario, which laptop model is the most suitable choice under the established criteria and weights when ranked with TOPSIS?

## 1.5 Scope and Structure of the Report

The unit of analysis is a laptop considered for office work by a Vietnamese buyer. The study excludes specialised gaming and graphics-workstation requirements and does not claim to represent all user segments or retail channels. The twelve alternatives originate from current recommendations published by seven professional review outlets; their price and configuration data are taken from Amazon.com because most of the newly released international models were not available in the Vietnamese catalogues reviewed at the collection date. Market data are therefore treated as a dated snapshot rather than a stable property of the alternatives. The study further uses a single representative expert for AHP weighting, so the resulting priority vector reflects one coherent expert judgement rather than a group-AHP consensus.

The remainder of the report is organised into five chapters. Chapter 2 establishes the theoretical foundations of MCDM, AHP, and TOPSIS and synthesises prior laptop and electronics

selection studies. Chapter 3 explains the staged criteria-development process, preference elicitation, alternative identification, and computational procedures. Chapter 4 reports the screening, weighting, and ranking results. Chapter 5 interprets those findings against prior evidence, and Chapter 6 summarises the contributions, limitations, and directions for further research.

# Chapter 2

## Literature Review

### 2.1 Laptop Selection as a Multi-Criteria Decision Problem

Choosing a laptop requires weighing attributes that cannot be reduced to a single scale. Processing performance, battery life, durability, portability, and price move in different directions and often trade off against one another, since a lighter chassis tends to shrink battery capacity and a lower price tends to constrain processing power. Sönmez Çakır and Pekkaya (2020) frame this directly, noting that new features are added to laptops every year and that the resulting variety of brands, models, and configurations forces buyers into a comparison problem that intuition alone cannot resolve consistently. Gaur (2023) reach a similar conclusion from the demand side, arguing that as shipments grew sharply during the pandemic, the number of criteria buyers had to reconcile, from brand and price to processor and screen size, grew with it. A decision problem of this kind, where several conflicting criteria must be evaluated jointly against a set of alternatives, falls within the scope of Multi-Criteria Decision-Making (MCDM), the methodological family this chapter now turns to.

### 2.2 An Overview of MCDM Methods

MCDM research has grown into an established branch of operations research. Based on a Scopus bibliometric analysis, Zyoud and Fuchs-Hanusch (2017) report 10,188 documents on AHP and 2,412 documents on TOPSIS through 2015, with China producing the largest national share of each (34.5% and 35.1%, respectively) and *Expert Systems with Applications* standing out as the most productive outlet for both techniques. This scale of activity reflects a broad family of methods rather than a single dominant technique.

As an organising lens, this study adopts the five-group taxonomy used by Villalba et al. (2024) in their domain review and reads it alongside the broader cross-domain evidence assembled by Basílio et al. (2022). The five families, laid out in Figure 2.1, differ less in the problems they address than in the arithmetic they use to reach a ranking. At one end, direct scoring meth-



ods such as Simple Additive Weighting and its extension COPRAS aggregate weighted performance scores into a single index, and the utility and value family, including MAUT, MAVT, and MIVES, does much the same after first routing each criterion through an explicit utility function. Distance-based methods, principally TOPSIS and VIKOR, take a geometric view instead, ranking alternatives by how close they lie to an ideal solution, whereas pairwise comparison methods led by AHP, alongside the Analytic Network Process and MACBETH, work from relative judgements between pairs rather than from absolute scores at all. Outranking methods such as PROMETHEE and ELECTRE occupy a different position again, building a preference relation from the degree to which one alternative outranks another rather than collapsing performance into a single number. This taxonomy is useful because it shows that AHP and TOPSIS belong to different groups, pairwise comparison and distance-based respectively, so that combining them addresses two distinct parts of the decision problem, weighting and ranking, rather than duplicating the same mechanism twice.

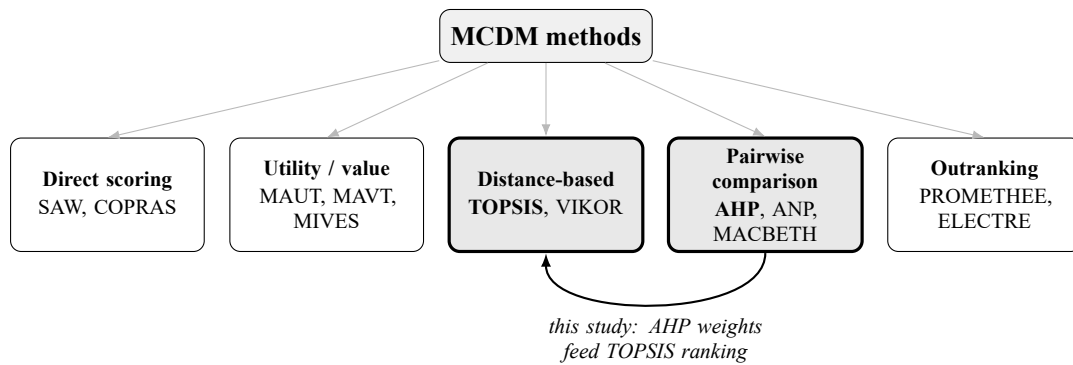


Figure 2.1: Families of MCDM methods.

*Note:* representative methods are listed under each family; AHP and TOPSIS, used in this study, are shown in bold.

*Source:* adapted from Villalba et al. (2024).

The prevalence of these two techniques is confirmed by counts of published applications. Basílio et al. (2022), reviewing 23,494 documents published between 1977 and 2022, record 6,835 applications of AHP and 4,907 of TOPSIS, ahead of VIKOR (1,475), PROMETHEE (1,382), ANP (1,262), ELECTRE (1,005), and DEMATEL (888). Villalba et al. (2024) report a comparable pattern within their own building-assessment domain, where the distance-based and pairwise comparison groups identified above are the two most frequently applied, AHP alone accounts for 35% of the reviewed articles and its combination with TOPSIS for a further 31%, ahead of the next most common technique, Simple Additive Weighting, at 11%, and AHP overall, whether used alone or hybridised with another technique, appears in 71% of the sample. Hybridisation with another MCDM method occurs in 55 of the reviewed articles, while only 5 propose a modification to an existing method rather than applying it as originally formulated, indicating that researchers in this domain favour combining established techniques over developing new ones. Tighnavard Balasbaneh et al. (2025) likewise observe that combining AHP

and TOPSIS is used more often than other method pairings, although they caution that no single combination has emerged as a settled consensus across application domains. Taken together, these figures indicate that AHP and TOPSIS are not only individually common but are also the pairing most frequently chosen when a study needs both criteria weights and an alternative ranking, which motivates a closer look at each method in turn.

## 2.3 The Analytic Hierarchy Process

The Analytic Hierarchy Process was introduced by Saaty (1987) as a method of measurement built on ratio scales derived from pairwise comparison. According to Zyoud and Fuchs-Hanusch (2017), AHP rests on three principles. Decomposition structures a decision into a hierarchy of a goal, criteria, and alternatives, illustrated in its complete form in Figure 2.2; comparative judgement elicits the relative importance of elements at the same hierarchical level through pairwise comparison; and synthesis of priorities aggregates these judgements into an overall priority vector.

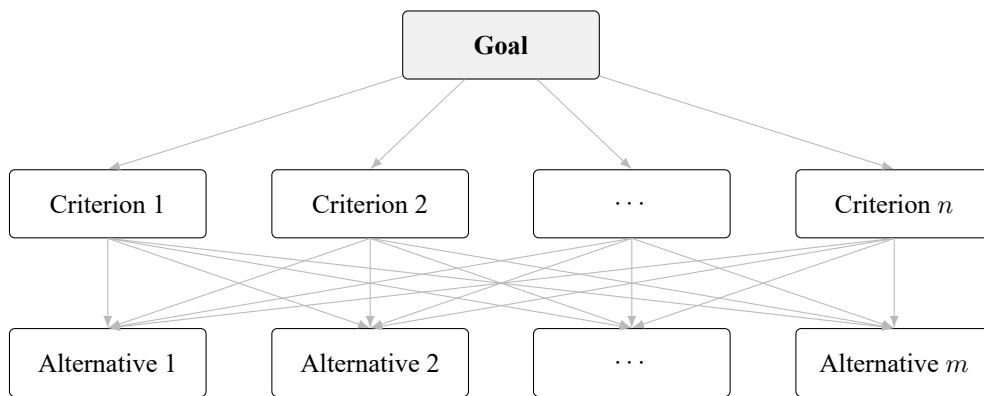


Figure 2.2: The complete three-level AHP hierarchy.

Source: adapted from Šostar & Ristanović (2023).

Comparisons are expressed on a nine-point ratio scale (Table 3.1), which allows both quantitative and qualitative criteria to be evaluated within the same framework, a property that makes AHP well suited to decisions such as laptop selection in which some attributes, for instance processor performance, are naturally numeric while others, such as brand perception, are not.

Because the judgements underlying a pairwise comparison matrix are elicited from a human respondent rather than measured directly, AHP incorporates an explicit check on their internal coherence. The consistency ratio, obtained from the principal eigenvalue of the comparison matrix as detailed in Section 3.3 (Equation (3.3)) and formalised by Saaty (2008), flags matrices in which the respondent's judgements contradict one another, for example by rating criterion A more important than B, B more important than C, yet C more important than A. This consistency check is one of the reasons AHP remains attractive for eliciting subjective expert weights, because unlike a simple ranking or scoring exercise it provides a quantitative signal for when

a judgement set should be revisited. The method is not without limitations, however. Zyoud and Fuchs-Hanusch (2017) note that AHP is susceptible to rank reversal, whereby the relative ranking of two existing alternatives can change when a new, unrelated alternative is added to the comparison, and that the pairwise comparison procedure assumes criteria are independent of one another, an assumption that does not always hold in practice.

Where expert judgements are too imprecise to be expressed as single crisp numbers, several studies extend AHP into a fuzzy environment. Liu et al. (2020) organise this literature around four technical choices that a fuzzy AHP model must make. It must decide how the relative importance of a comparison is represented, typically as a fuzzy number rather than a crisp value; how judgements from multiple experts or multiple criteria are aggregated; how the resulting fuzzy priorities are defuzzified back into crisp weights for practical use; and how consistency is measured once judgements are no longer single numbers. This layer of fuzzification is warranted when the study is designed to preserve linguistic uncertainty rather than require a single comparison value. The present study instead asks one representative expert to provide explicit Saaty-scale judgements and reports the resulting matrix in full. Crisp AHP is therefore selected for parsimony and auditability, not because uncertainty is assumed to be absent. This choice also bounds the interpretation, since the priority vector represents one internally consistent expert judgement and should not be read as a group consensus.

The independence assumption and the possibility of rank reversal remain relevant design limitations. The model treats the retained criteria as analytically separable even where practical relationships exist, such as those between battery capacity and weight. Unlike the complete hierarchy of Figure 2.2, AHP is used only to weight criteria rather than to rank the laptop alternatives, which reduces but does not eliminate the need to examine how the final ranking responds to the chosen criteria structure and weights.

## 2.4 The TOPSIS Method

The Technique for Order of Preference by Similarity to Ideal Solution was proposed by Hwang and Yoon (1981) on the principle that the best alternative should lie simultaneously closest to a positive ideal solution and farthest from a negative ideal solution, where the positive ideal solution combines the best attainable value on every criterion and the negative ideal solution combines the worst. Behzadian et al. (2012) note that this requires attribute values that are numeric, monotonically increasing or decreasing in desirability, and measured on commensurable scales, conditions that a laptop dataset of prices, benchmark scores, and physical measurements satisfies directly.

Reviewing 266 applications of TOPSIS published across 103 journals since 2000, Behzadian et al. (2012) find the method concentrated in supply chain management and logistics (27.5% of the sample) and in design, engineering, and manufacturing systems (23%), together accounting for more than half of the applications surveyed. The same review attributes this spread to several

practical advantages. TOPSIS makes full use of the available attribute information rather than discarding it at an intermediate step, produces a cardinal ranking of every alternative rather than a partial order, does not require the criteria to be preferentially independent, and remains conceptually and computationally simple regardless of how many criteria or alternatives are involved. Zyoud and Fuchs-Hanusch (2017) similarly place TOPSIS as the second most frequently used MCDM technique after AHP, consistent with the frequency counts reported in Section 2.2.

These advantages do not make a TOPSIS ranking invariant. The method is compensatory, in that strong performance on one criterion can offset weak performance on another once both enter the weighted distance calculation. Its result also depends on the normalisation rule, the weight vector, and the observed alternative set from which the ideal and anti-ideal values are constructed. A change to the candidate set can therefore alter the reference points and the relative closeness of existing alternatives. In this study, the response is procedural. The candidate set and collection date are fixed, one normalisation rule is used throughout, the raw and transformed decision matrices are disclosed, and small differences among the leading closeness coefficients are interpreted as a short list rather than as evidence of an absolute winner.

As with AHP, a fuzzy extension exists for settings in which attribute values themselves are imprecise rather than directly measurable. Chen (2000) extend TOPSIS to a fuzzy environment by representing ratings and weights as triangular fuzzy numbers, computing distances to a fuzzy positive and negative ideal solution through the vertex method, and deriving a closeness coefficient from these distances. This extension is most useful when performance data are themselves elicited as linguistic judgements. The laptop dataset used in this study, in contrast, is built from directly observable specifications and market prices, so the crisp TOPSIS procedure described in Section 3.4 is applied without a fuzzy layer.

## **2.5 Prior Studies on Laptop and Electronics Selection**

Applying MCDM to laptop and related electronics choices is an established but fragmented line of research, summarised in Table 2.1.

Table 2.1: Prior studies on laptop and electronics selection using MCDM.

| Source                        | Context / data                        | Criteria input                                    | Weighting / ranking                    |
|-------------------------------|---------------------------------------|---------------------------------------------------|----------------------------------------|
| Sönmez Çakır & Pekkaya (2020) | Laptop; 23 experts                    | Literature + expert refinement                    | Fuzzy AHP; DEMATEL; no product ranking |
| Gaur (2023)                   | Nine laptop brands; pandemic setting  | Nine technical and market criteria                | AHP + BWM weights; TOPSIS ranking      |
| Weng Siew Lam et al. (2023)   | Mobile-phone alternatives             | Technical, physical, and user-related criteria    | AHP weights; TOPSIS ranking            |
| Maghsoudi et al. (2026)       | Notebook reviews and expert knowledge | Review-derived product features                   | Sentiment + fuzzy weighting + TOPSIS   |
| Rau & Fang (2018)             | Notebook purchase timing              | Value, brand, cost, and eco-efficiency            | Scenario weights; TOPSIS ranking       |
| Liao & Chuang (2022)          | Eco-friendly laptop experiment        | Price, battery, shell, CPU, and display           | Conjoint analysis                      |
| Kang et al. (2022)            | Online laptop reviews                 | Product, logistics, and after-sales attributes    | Empirical review mining                |
| Guan & Yuen (2022)            | Personalised laptop recommendation    | Product attributes and user preferences           | Pairwise rating + clustering           |
| Pramanik et al. (2021)        | Mobile computing resources            | Computing-resource parameters                     | Comparative MCDM ranking               |
| Šostar & Ristanović (2023)    | General consumer behaviour            | Cultural, social, personal, psychological factors | AHP weighting                          |
| Jiménez-Parra et al. (2014)   | Remanufactured laptops in Spain       | Attitude, norms, and motivation                   | Behavioural model                      |

Source: Authors.

The evidence in Table 2.1 shows agreement on the multidimensional character of the decision but disagreement on which dimension should dominate. Sönmez Çakır and Pekkaya (2020) place brand image first and show that brand, price, and peripheral properties account for roughly two-thirds of total priority. By contrast, Liao and Chuang (2022) identify price as the most influential attribute in an eco-friendly laptop experiment, while Weng Siew Lam et al. (2023) assign the greatest technical importance to battery life in their mobile-phone model. These differences are not measurement noise alone; they reflect different products, populations, criteria definitions, and elicitation procedures.

The studies also confirm that integrated ranking is not methodologically novel by itself. Gaur (2023) combine AHP and BWM weights with TOPSIS to rank laptop brands, and Rau and Fang (2018) use TOPSIS with real notebook data to rank purchase options over time. Related work extends the evidence base beyond MCDM. Kang et al. (2022) show that online purchase decisions incorporate logistics and after-sales experience alongside product configuration, while Jiménez-Parra et al. (2014) demonstrate that attitudes and social norms matter in the remanufactured-laptop context. A technically complete ranking can therefore remain contextually incomplete if its criteria are fixed before the intended users and local procurement conditions are examined.

## 2.6 Research Gap and Positioning

The gap is consequently located in evidence integration rather than in the invention of a new hybrid method. Existing laptop-ranking studies demonstrate that AHP, BWM, fuzzy weighting, and TOPSIS can be combined, but they generally begin with criteria selected for a different user group or study context. Consumer studies identify additional economic, experiential, and service attributes but do not carry them through to an auditable alternative ranking. The present study links these streams by separating four decisions that prior work often compresses, namely which criteria enter the candidate set, which survive screening by the target population, how their relative weights are elicited, and how a fixed alternative snapshot is ranked.

This positioning also defines what the study does not claim. It applies an established AHP–TOPSIS pairing rather than proposing a new algorithm, and it evaluates internationally recommended models priced through Amazon.com rather than claiming coverage of the Vietnamese retail market. Its contribution lies in the traceable calibration of the criteria to Vietnamese office work and in the disclosure of the evidence and calculations connecting that calibration to the final ranking.

# Chapter 3

## Methodology

### 3.1 The Proposed Research Model

The study is organised into three sequential stages, each of which answers one of the research questions stated in Section 1.4. The first stage develops the set of evaluation criteria. Candidate criteria are synthesised from the literature reviewed in Chapter 2, refined through expert interviews, and then screened through a convenience survey of respondents rating each criterion for office-work relevance, so that only criteria receiving broad support beyond the expert panel enter the quantitative model. The second stage quantifies the relative importance of the retained criteria with AHP, including an explicit consistency check before the weights are accepted. The third stage applies the model to a concrete purchase scenario: candidate laptops are identified from the recommendation lists of professional review outlets, their price and configuration data are collected from Amazon.com at a recorded date and operationalised into a decision matrix, and TOPSIS ranks them using the weights obtained in the second stage. The stages are strictly ordered, since the pairwise comparison in Stage 2 cannot be designed until the criteria set is fixed, and the ranking in Stage 3 is meaningful only after the weights have passed the consistency check. Figure 3.1 summarises this pipeline.

### 3.2 Stage 1 — Criteria Development

#### 3.2.1 Literature Synthesis

The starting point of the criteria set is the body of laptop-selection and consumer-behaviour studies reviewed in Section 2.5. From these sources, every attribute that at least one prior study used to evaluate laptops or closely related devices was extracted and consolidated. Price appears in nearly all of them (Liao & Chuang, 2022; Rau & Fang, 2018; Sönmez Çakır & Pekkaya, 2020; Weng Siew Lam et al., 2023), core hardware attributes such as processor, memory, and storage recur across the MCDM applications (Gaur, 2023; Maghsoudi et al., 2026), and service-related attributes such as brand and after-sales support are emphasised in the review-mining

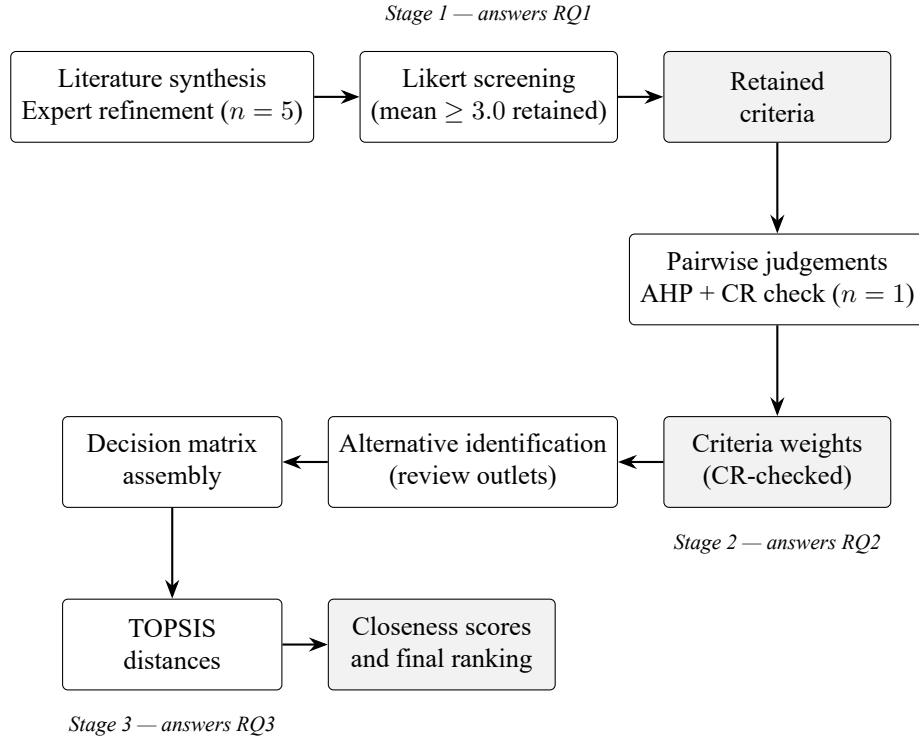


Figure 3.1: The proposed research model.

Source: Authors.

and consumer-behaviour literature (Kang et al., 2022; Šostar & Ristanović, 2023). Attributes that overlap in meaning were merged into a single criterion, and attributes tied to specialised use cases outside the office-worker persona, such as colour-accurate displays for design work, were excluded at this stage. The remaining attributes were then organised into thematic groups reflecting different aspects of the purchase decision, such as core hardware capability or trust and after-sales value, purely to aid the presentation and screening of the candidate set in Table 4.1; the criteria that survive screening are compared through a single flat pairwise matrix in Section 3.3 rather than through a grouped or multi-level AHP hierarchy. The resulting set of candidate criteria is reported in Section 4.1.1.

### 3.2.2 Expert Interviews

Because the candidate criteria from Section 3.2.1 were derived from studies conducted outside Vietnam, they were validated against local practice before being quantified. Five experts were selected purposively so that each occupies a different position in the laptop procurement process. They comprise a deputy head of an international relations office with seven years of end-user experience (CG-01), an IT administrator who specifies and maintains office fleets (CG-02), an import–export documentation officer representing the target persona (CG-03), a head of product development with ten years of managerial purchasing experience (CG-04), and a retail sales consultant who advises laptop buyers daily (CG-05).



The interview protocol itself, rather than being improvised, was informed by the Interview Protocol Refinement (IPR) framework, whose four phases call for every question to be traceable to a research objective, for the questions to be sequenced as an inquiry-based conversation rather than a checklist, for the draft to be reviewed before fieldwork, and for the instrument to be piloted (Castillo-Montoya, 2016). The present protocol follows the first two phases in full and was reviewed internally before fieldwork, though no separate pilot interview was conducted ahead of the five recorded sessions. Applying this framework produced a five-part funnel that moves from the concrete to the general, comprising a warm-up section grounding each respondent in a recent, specific laptop-selection episode; a stimulus-based section in which a literature finding for each criterion group is read aloud immediately before the matching question, so that agreement or disagreement is anchored to a stated claim rather than solicited as an abstract opinion; an open section probing for criteria specific to the Vietnamese context that the literature synthesis could not have anticipated; a trade-off scenario that generates qualitative seed data for the pairwise comparisons of Section 3.3; and a closing section built around member-checking, in which the interviewer restates the respondent's positions for confirmation. Most substantive questions in this structure carry a follow-up probe that asks the respondent to recall the specific case behind an answer or to explain the reasoning process that produced it, a technique adapted from cognitive interviewing that keeps a response anchored in the respondent's actual experience rather than a general impression formed on the spot (Willis, 2005). This funnel structure and probing technique operate inside a semi-structured session that functions as a structured expert refinement stage preceding AHP, an arrangement applied in prior criteria-development work (Al Hazza et al., 2022; McMillan et al., 2016). Each session lasted twenty to thirty minutes; all five respondents gave informed consent, no audio was recorded, and only written notes were retained.

Interview notes were reviewed for proposals to add, remove, or redefine a criterion. To keep the revision process auditable rather than ad hoc, a proposal was adopted only when it was corroborated independently by a majority of the five experts, and any resulting change to the candidate set was cross-checked against the literature reviewed in Section 2.5 before being finalised. The outcome of this revision process is reported together with the resulting criteria set in Section 4.1.1.

### **3.2.3 Survey-Based Screening**

Expert validation establishes that a criterion is defensible, but not that the target population actually weighs it when buying. The candidate criteria from Section 3.2.2 were therefore submitted to a wider screening survey before any pairwise comparison was designed. A questionnaire was distributed online through Google Forms as a self-administered, voluntary-response instrument, open to any respondent willing to complete it rather than restricted by a formal occupational eligibility check; respondents rated the importance of each criterion when choosing a laptop for

office work on a five-point Likert scale, from 1 (not important) to 5 (very important). The instrument deliberately contained one rating item per candidate criterion and no other items, keeping completion time short and avoiding demographic questions that the screening decision does not use. Because each criterion is measured by a single item rather than a multi-item scale, the exercise functions as item-level importance screening rather than validation of a latent psychometric construct, and because the sample was recruited by convenience rather than by a method targeting demographically verified office workers, the 84 responses collected are best read as a screening signal from a broadly relevant respondent pool rather than as a confirmed office-worker panel; this qualification carries through the reporting of results in Section 4.2 and is revisited as a limitation in Section 6.3.

The retention rule was fixed before data collection. A criterion is retained if its sample mean is at least 3.0, the midpoint of the scale, and removed otherwise; this is a pragmatic decision rule keyed to the scale midpoint rather than a psychometrically validated cut-off. On a five-point scale the midpoint separates responses that lean towards importance from those that lean away from it, so a mean below 3.0 indicates that respondents on balance do not regard the criterion as important (Chyung et al., 2017). Committing to the threshold in advance prevents the rule from being adjusted after the results are known. Responses missing more than 20% of the items were to be discarded, and straight-lined responses, in which a respondent gives the same rating to every item, were to be flagged for a group decision rather than silently removed. The screening outcome, which reduces the candidate set to the criteria carried into Stage 2, is reported in Section 4.2.

### 3.3 Stage 2 — AHP for Criteria Weighting

The relative importance of the criteria retained after screening is derived through the Analytic Hierarchy Process, whose theoretical basis is reviewed in Section 2.3. The procedure follows five steps.

**Step 1 — Hierarchy construction.** The AHP component of the model is structured into two levels only, the goal of selecting the most suitable laptop for a Vietnamese office worker at the top and the seven criteria retained from Stage 1 immediately below it. The candidate laptop models are not entered into this hierarchy and are not compared through pairwise judgement at any point. Alternatives are instead introduced in Stage 3, where TOPSIS evaluates them directly on measurable attribute data using the weights this stage produces.

**Step 2 — Pairwise comparison.** The pairwise judgements were elicited from one representative expert drawn from the interview panel of Section 3.2.2, selected for direct, first-hand experience of office laptop procurement. For  $n$  retained criteria the expert completes  $n(n - 1)/2$  comparisons, each expressing how much more important one criterion is than another on the

nine-point ratio scale of Table 3.1. Relying on a single judgement set concentrates the weighting in one perspective, which is why the consistency check in Step 5 is applied strictly. An inconsistent matrix is returned to the expert for re-evaluation rather than accepted or averaged away.

Table 3.1: The Saaty fundamental scale of pairwise comparison.

| Intensity  | Definition             |
|------------|------------------------|
| 1          | Equal importance       |
| 3          | Moderate importance    |
| 5          | Strong importance      |
| 7          | Very strong importance |
| 9          | Extreme importance     |
| 2, 4, 6, 8 | Intermediate values    |

*Source:* adapted from Saaty (1987).

**Step 3 — Comparison matrix.** For  $n$  criteria, the pairwise judgements form a reciprocal matrix  $A = [a_{ij}]$  with  $a_{ij} = 1/a_{ji}$  and  $a_{ii} = 1$ :

$$A = \begin{bmatrix} 1 & a_{12} & \cdots & a_{1n} \\ 1/a_{12} & 1 & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 1/a_{1n} & 1/a_{2n} & \cdots & 1 \end{bmatrix}. \quad (3.1)$$

Only the  $n(n-1)/2$  entries above the diagonal are elicited; the remaining entries follow from reciprocity, so the expert is never asked the same comparison twice.

**Step 4 — Priority (weight) vector.** The criteria weights are taken as the principal right eigenvector of  $A$ , that is, the vector  $\mathbf{w}$  satisfying

$$A\mathbf{w} = \lambda_{\max} \mathbf{w}, \quad \sum_{i=1}^n w_i = 1, \quad (3.2)$$

where  $\lambda_{\max}$  is the largest eigenvalue of  $A$ . Several simpler approximations circulate in applied work, including the normalised column average used by Weng Siew Lam et al. (2023); the eigenvector is preferred here because Saaty (2003) shows it to be the theoretically required solution whenever the judgement matrix is a small perturbation of a fully consistent one, which the consistency check in Step 5 verifies, and because the comparative simulations of Ishizaka and Lusti (2006) find that approximation methods diverge from the eigenvector increasingly as matrices grow larger or less consistent. In computation the eigenvector is obtained by power iteration, repeatedly multiplying a normalised weight vector by  $A$  until convergence. As a cross-check, the weights are also computed by the geometric mean of each row; agreement between

the two vectors in the ordering of criteria shows that the priority ordering is stable across these two computational routes, though it does not by itself establish that the ordering would hold under a different elicitation or aggregation method more broadly.

**Step 5 — Consistency check.** Because the judgements are elicited from a human respondent, they may contradict one another, and the eigenvector of a heavily contradictory matrix is not a meaningful weight vector. The internal coherence of the judgements is therefore verified through the consistency ratio  $CR = CI/RI$ , where the consistency index is

$$CI = \frac{\lambda_{\max} - n}{n - 1}, \quad (3.3)$$

$\lambda_{\max}$  is the principal eigenvalue from Equation (3.2), and  $RI$  is the random index, the mean consistency index of randomly generated reciprocal matrices of the same order (Table 3.2). A matrix is accepted when  $CR \leq 0.10$  (Saaty, 2008). If the threshold is exceeded, the weights are not used. The three pairwise entries that deviate most from the values a consistent matrix would imply are identified and returned to the expert for re-evaluation, and the matrix is recomputed. The threshold is never relaxed to accommodate a marginally failing matrix.

Table 3.2: Random Index (RI) values by matrix order  $n$ .

| $n$ | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   |
|-----|------|------|------|------|------|------|------|------|
| RI  | 0.58 | 0.90 | 1.12 | 1.24 | 1.32 | 1.41 | 1.45 | 1.49 |

*Source:* adapted from Saaty (1987).

## 3.4 Stage 3 — Scenario-Based Application with TOPSIS

### 3.4.1 Decision Scenario

The third stage applies the weighted criteria to a concrete decision situation. The scenario mirrors the problem statement of Section 1.2, in which an office worker in Vietnam needs a laptop for work, has browsed review websites and the catalogues of established retailers, and still cannot decide, because the information available compares products one attribute at a time while the actual decision requires weighing all attributes at once. Framing Stage 3 as a scenario serves two purposes. It keeps the demonstration honest, since every modelling choice, from which laptops enter the comparison to how each criterion is measured, must be one this buyer could reproduce from publicly available information; and it makes the output directly interpretable, since the resulting ranking answers the question the buyer actually asks. The stage proceeds in three steps that parallel a standard data workflow, namely identifying the alternatives, collecting and operationalising their attribute data, and analysing the resulting decision matrix with TOPSIS.

### 3.4.2 Alternative Identification

The candidate laptops are not chosen by the researchers’ own judgement, which would risk pre-selecting the winner, but harvested from published recommendation lists of professional review outlets and then filtered for local availability. An outlet is included in the source pool if it satisfies three conditions. It must maintain a documented, standardised laboratory testing protocol rather than opinion-only reviews, actively cover laptops as a product category, and keep an editorial operation stable enough that its current recommendations reflect that protocol. Seven outlets meet these conditions, summarised in Table 3.3. One widely known outlet, Engadget, was considered and excluded, because it reduced its editorial staff by roughly one third in 2024, reoriented its output towards search-driven affiliate content, and changed ownership in early 2026, so the stability condition is not met. CNET is retained with a caveat. The outlet published AI-generated articles without clear disclosure in 2023, but it operates its own testing laboratory and tightened human editorial review after the incident, and its current laptop recommendations are lab-based.

Table 3.3: Review outlets used as sources for alternative identification.

| Outlet      | Operator           | Testing approach                                                  |
|-------------|--------------------|-------------------------------------------------------------------|
| PCMag       | Ziff Davis         | Standardised lab benchmarks since 1982; Editors’ Choice programme |
| Tom’s Guide | Future plc         | Comparative product testing; seasonal category rankings           |
| TechRadar   | Future plc         | Global review team; per-segment rankings (business, budget)       |
| Laptop Mag  | Future plc         | Laptop-only coverage; in-house battery and performance benchmarks |
| CNET        | Ziff Davis         | In-house lab for performance, battery, and display testing        |
| Wirecutter  | The New York Times | Hands-on testing; no sponsored review placements                  |
| RTINGS      | Independent        | Instrumented measurement (display, thermals, input latency)       |

*Source:* Authors.

From each outlet in the pool, the current recommendation list for the best business or premium laptops of the model year is collected, and the union of these lists forms the raw candidate set. Because the same model frequently appears on several lists, the set is deduplicated by manufacturer and model designation, treating configuration variants of the same model as a single candidate at this step. Price and configuration data for the deduplicated set are then collected from Amazon.com rather than from a Vietnamese retailer, because the outlets’ recommendations are dominated by newly launched, internationally released configurations that Vietnamese retail catalogues do not yet carry at the time of collection, so the buyer of the scenario in Section 3.4.1 would in practice have to source most of these models through an international retailer regardless of the Vietnamese market context. Amazon.com is the world’s largest online retailer

and the leading e-commerce platform in the United States, and using its product listings as a source of laptop specification and price data has direct precedent in the consumer-electronics literature, where Rau and Fang (2018) construct their notebook price-history dataset directly from Amazon product listings. The date of collection is recorded with the data, since both the recommendation lists and the retailer's catalogue change over time.

### **3.4.3 Data Collection and Operationalisation**

Each remaining alternative is scored on the retained criteria, and the way each criterion is measured is fixed before the data are collected. Five criteria are directly quantitative and are taken as listed on Amazon.com, namely the price in US dollars, the memory capacity in gigabytes, the storage capacity in gigabytes, the device weight in kilograms (converted from the pounds figure shown in the listing), and the battery capacity in watt-hours as specified by the manufacturer. Watt-hours are used in preference to a manufacturer-claimed runtime in hours, because the latter is measured under differing, vendor-specific workloads and is therefore not commensurable across brands, whereas watt-hour capacity is a fixed physical specification of the battery itself. The processor cannot be treated this way, because chipset designations such as product names and generation labels are neither numeric nor directly comparable across manufacturers. The criterion is therefore operationalised through a standardised benchmark score, using the Pass-Mark CPU database, which converts heterogeneous processor names into a single performance scale. This step matters for the validity of the analysis, because TOPSIS requires attribute values that are numeric, monotonic in desirability, and commensurable (Behzadian et al., 2012); a benchmark score satisfies all three conditions where a chipset name satisfies none. Brand, the one qualitative criterion, is likewise converted into an observable quantity rather than scored by hand. It is operationalised as each manufacturer's worldwide PC shipment market share for the first quarter of 2026, as reported by the market-research firm Gartner (Gartner, Inc., 2026). This proxy is chosen in preference to a Vietnam-specific measure, such as a count of authorised service centres, because no freely accessible source publishes brand-level market share for the Vietnamese laptop market specifically; global shipment share is disclosed quarterly and is independently reproducible, and it still functions as the concrete, verifiable expression of manufacturer standing that the trust and after-sales group of Table 4.1 calls for, whereas a hand-assigned brand score would reintroduce exactly the subjectivity that the screening and weighting stages were designed to control. Gartner is used in preference to IDC, the other major PC-market tracker, because Gartner's quarterly release reports a share figure for every one of the six brands present in this candidate set, including Acer, whereas IDC's release names only its top five vendors by shipment volume and groups every remaining brand, Acer included, into a single undifferentiated "other vendors" residual; using a single source that individually reports all six brands avoids mixing two trackers' figures within the same criterion column.

Table 3.4: Operational definitions of the TOPSIS criteria.

| Criterion     | Operational measure                     | Unit   | Type    | Primary source                |
|---------------|-----------------------------------------|--------|---------|-------------------------------|
| Price         | Listed purchase price                   | USD    | Cost    | Amazon.com listing            |
| CPU           | PassMark CPU benchmark score            | Points | Benefit | PassMark CPU database         |
| RAM           | Installed memory capacity               | GB     | Benefit | Amazon.com listing            |
| SSD           | Solid-state storage capacity            | GB     | Benefit | Amazon.com listing            |
| Brand         | Worldwide PC shipment share, Q1 2026    | %      | Benefit | Gartner (Gartner, Inc., 2026) |
| Size / weight | Device mass converted to kilograms      | kg     | Cost    | Amazon.com listing            |
| Battery life  | Manufacturer-specified battery capacity | Wh     | Benefit | Product specification         |

*Source: Authors.*

### 3.4.4 The TOPSIS Procedure

Given the criteria weights from Stage 2 and the decision matrix assembled above, the candidate laptops are ranked with TOPSIS over seven steps.

**Step 1 — Decision matrix.** For  $m$  alternatives and  $n$  criteria,  $X = [x_{ij}]$  collects the performance of alternative  $i$  on criterion  $j$ .

**Step 2 — Vector normalisation.**

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}}. \quad (3.4)$$

**Step 3 — Weighted normalised matrix.**

$$v_{ij} = w_j r_{ij}. \quad (3.5)$$

**Step 4 — Ideal solutions.** The positive ideal solution  $A^+$  and negative ideal solution  $A^-$  are formed from the best and worst weighted scores of each criterion, respecting its benefit or cost direction:

$$A^+ = \{v_1^+, \dots, v_n^+\}, \quad A^- = \{v_1^-, \dots, v_n^-\}. \quad (3.6)$$

**Step 5 — Separation measures.**

$$S_i^+ = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^+)^2}, \quad S_i^- = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^-)^2}. \quad (3.7)$$

**Step 6 — Closeness coefficient.**

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}, \quad 0 \leq C_i \leq 1. \quad (3.8)$$

**Step 7 — Ranking.** The alternatives are ranked in descending order of  $C_i$ ; the laptop with the highest  $C_i$  is the recommended choice.



# Chapter 4

## Results

### 4.1 Criteria Development Outcomes

#### 4.1.1 The Candidate Criteria Set

Applying the literature-synthesis procedure of Section 3.2.1 to the studies reviewed in Section 2.5 produced fourteen candidate criteria. These were organised into six groups reflecting the aspects of the purchase decision they describe: economic cost, core hardware capability, design and interaction experience, portability and durability, ecosystem utility, and trust and after-sales value. Table 4.1 lists the resulting set together with each criterion’s group and its direction of preference. Most criteria are benefit criteria, for which a larger value is preferred; price and physical size or weight are cost criteria, for which a smaller value is preferred. The distinction matters at two later points. It determines the direction of the ideal solutions in TOPSIS (Section 3.4), and it guards against the error of treating a heavier or more expensive machine as better simply because its raw attribute value is larger.

#### 4.1.2 Interview-Driven Refinement

Following the revision rule described in Section 3.2.2, the interviews changed the candidate set in two ways. First, the dedicated graphics card was removed. Four of the five experts converged independently on the argument that a discrete GPU is a cost without a corresponding benefit for office work, since it raises the purchase price, weight, and thermal load while shortening battery life, and the budget it consumes is better redirected towards a newer-generation processor, more memory, or faster storage. This judgement is consistent with the finding of Maghsoudi et al. (2026) that the graphics card ranked last among ten notebook attributes in their expert-weighted model. Second, software compatibility was added. The experts who work inside Vietnamese offices (CG-01, CG-03) described a lock-in effect that the international literature rarely makes explicit, in which accounting packages, internal ERP systems, and the government’s customs declaration software are written for Windows, so a device that cannot run them is excluded

Table 4.1: The fourteen candidate evaluation criteria in six groups.

| ID  | Criterion               | Type    | Group                    |
|-----|-------------------------|---------|--------------------------|
| C1  | Price                   | Cost    | Economic                 |
| C2  | CPU                     | Benefit | Core hardware            |
| C3  | RAM                     | Benefit | Core hardware            |
| C4  | Storage (SSD)           | Benefit | Core hardware            |
| C5  | Display                 | Benefit | Design & interaction     |
| C6  | Keyboard                | Benefit | Design & interaction     |
| C7  | Design & shell material | Benefit | Design & interaction     |
| C8  | Size / weight           | Cost    | Portability & durability |
| C9  | Durability              | Benefit | Portability & durability |
| C10 | Battery life            | Benefit | Portability & durability |
| C11 | Connectivity ports      | Benefit | Ecosystem utility        |
| C12 | Software compatibility  | Benefit | Ecosystem utility        |
| C13 | Brand                   | Benefit | Trust & after-sales      |
| C14 | After-sales / warranty  | Benefit | Trust & after-sales      |

Source: Authors.

from consideration regardless of its other merits. The interviews also sharpened the definitions of two retained criteria, restricting the storage criterion to solid-state drives and extending the design criterion to cover shell material, and recalibrated the expectation for battery life, because mains power is readily available in offices, so a runtime of four to seven hours was considered sufficient, a markedly lower bar than the top-priority status that battery life receives in Weng Siew Lam et al. (2023). The revised set therefore still contains fourteen criteria, but with GPU replaced by software compatibility and with definitions anchored to local working conditions; this is the candidate set reported in Table 4.1 above and carried into the screening survey of Section 3.2.3.

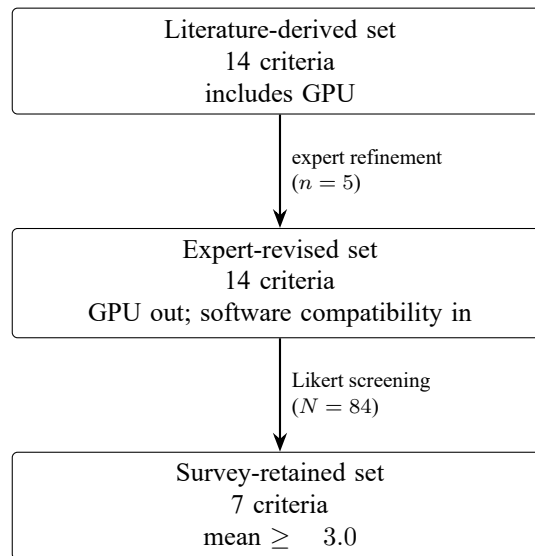


Figure 4.1: Transition from the literature-derived criteria to the final retained set.

Source: Authors.

## 4.2 Criteria Screening Results

The screening survey described in Section 3.2.3 collected 84 responses, all of which were complete. No respondent left more than 20% of the items unanswered, and no straight-lined response was detected, so every questionnaire entered the analysis. Table 4.2 reports the descriptive statistics for the fourteen candidate criteria together with the retention decision under the pre-committed threshold of a mean of at least 3.0.

Table 4.2: Descriptive statistics and screening decision for the candidate criteria.

| Criterion               | <i>N</i> | Mean | SD   | Decision |
|-------------------------|----------|------|------|----------|
| Price                   | 84       | 3.79 | 1.27 | Retain   |
| CPU                     | 84       | 3.86 | 1.13 | Retain   |
| RAM                     | 84       | 3.96 | 1.02 | Retain   |
| SSD                     | 84       | 3.89 | 1.12 | Retain   |
| Display                 | 84       | 2.80 | 1.32 | Remove   |
| Keyboard                | 84       | 2.88 | 1.07 | Remove   |
| Design & shell material | 84       | 2.83 | 1.16 | Remove   |
| Size / weight           | 84       | 3.96 | 1.02 | Retain   |
| Durability              | 84       | 2.36 | 1.42 | Remove   |
| Battery life            | 84       | 4.02 | 0.97 | Retain   |
| Connectivity ports      | 84       | 2.67 | 1.43 | Remove   |
| Software compatibility  | 84       | 2.54 | 1.43 | Remove   |
| Brand                   | 84       | 3.79 | 1.02 | Retain   |
| After-sales / warranty  | 84       | 2.37 | 1.34 | Remove   |

*Source:* Authors.

Seven criteria pass the threshold. Battery life records the highest mean (4.02), followed by memory and physical size or weight (both 3.96), storage (3.89), processor (3.86), and price and brand (both 3.79). The seven removed criteria all fall clearly below the midpoint, from keyboard experience (2.88) down to durability (2.36). The gap between the lowest retained mean (3.79) and the highest removed mean (2.88) spans almost a full scale point, so the composition of the retained set is not sensitive to small movements of the threshold around the midpoint. These seven criteria constitute the answer to RQ1 and define the dimension of the pairwise comparison in Stage 2. It is worth recording, ahead of the fuller treatment in Chapter 5, that several criteria emphasised in the expert interviews, notably software compatibility, durability, and after-sales support, were screened out by the wider respondent pool, a divergence between the expert panel and the survey respondents that the two-step design of Stage 1 makes visible rather than concealing.

### 4.3 Criteria Weights and Consistency Check

The retained criteria were compared pairwise by the representative expert as described in Section 3.3, producing a  $7 \times 7$  reciprocal matrix from 21 elicited judgements. Before the weights were interpreted, the matrix was submitted to the consistency check of Equation (3.3). The principal eigenvalue is  $\lambda_{\max} = 7.3823$ , giving a consistency index of  $CI = 0.0637$  and, with the random index  $RI = 1.32$  for  $n = 7$ , a consistency ratio of  $CR = 0.0483$ . As this value lies below the acceptance threshold of 0.10 (Saaty, 2008), the judgement set is internally coherent and the derived weights are usable without re-elicitation. Table 4.3 presents the resulting priority vector.

Table 4.3: Criteria weights obtained by AHP.

| Criterion     | Weight (eigenvector) |
|---------------|----------------------|
| Price         | 0.3491               |
| CPU           | 0.2769               |
| RAM           | 0.1201               |
| SSD           | 0.1592               |
| Brand         | 0.0430               |
| Size / weight | 0.0258               |
| Battery life  | 0.0258               |

Source: Authors.

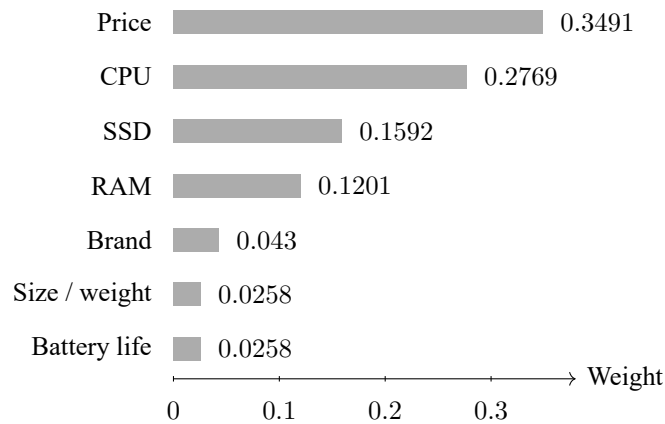


Figure 4.2: AHP weights of the seven retained criteria.

Source: Authors.

The weight structure is strongly concentrated at the top. Price carries the largest weight (0.3491) and the processor the second largest (0.2769); together these two criteria account for approximately 62.6% of the total priority, so the model characterises the office-laptop decision primarily as a trade-off between budget and computing capability. A middle tier follows, comprising storage (0.1592) and memory (0.1201), while brand (0.0430), size or weight (0.0258), and battery life (0.0258) form a low-priority tier that collectively holds less than 10% of the

weight. The position of battery life is notable, because although it recorded the highest mean in the screening survey, it receives the smallest AHP weight, and the same reversal separates this result from Weng Siew Lam et al. (2023), who report battery life as the most heavily weighted criterion in their mobile-phone model. This pattern is consistent with the recalibration that emerged from the expert interviews, in which battery life is a requirement that most office laptops already satisfy rather than a differentiator; its interpretation is developed in Chapter 5. As a robustness check, the weights were recomputed by the geometric-mean method described in Section 3.3; the resulting vector preserves the same ordering of all seven criteria, indicating that this ordering is stable across the two computational methods applied here rather than an artefact of the eigenvector procedure specifically. These weights answer RQ2 and enter Stage 3 as the weighting vector for TOPSIS.

## 4.4 Decision Scenario and Alternatives

Applying the outlet pool and deduplication procedure of Section 3.4.2 to the seven review outlets of Table 3.3 produced twelve distinct candidate laptops after configuration variants of the same model were collapsed into a single entry. Price and configuration data for all twelve candidates were collected from Amazon.com on 2026-07-12, following the sourcing rationale of Section 3.4.2. To keep the ranking table and the calculation tables of Chapter E compact, each candidate is assigned a short identifier, Can1 through Can12, in the order of collection; Table 4.4 reports the assignment, and the subsequent tables refer to the candidates by these identifiers.

Table 4.4: Candidate identifiers assigned to the twelve laptop models.

| Candidate | Model                                         |
|-----------|-----------------------------------------------|
| Can1      | Lenovo ThinkPad X1 Carbon Gen 13 Aura Edition |
| Can2      | Lenovo ThinkPad X1 Carbon Gen 14 Aura Edition |
| Can3      | Apple MacBook Air M4                          |
| Can4      | Apple MacBook Air M5                          |
| Can5      | Asus ExpertBook P5                            |
| Can6      | HP EliteBook Ultra G1i                        |
| Can7      | Lenovo ThinkPad X9 15 Aura Edition (2025)     |
| Can8      | Dell 14 Premium (XPS 14 2026)                 |
| Can9      | Lenovo ThinkPad T14 Gen 6 (AMD)               |
| Can10     | Lenovo ThinkPad T14 Gen 7                     |
| Can11     | Acer Aspire 5                                 |
| Can12     | Lenovo ThinkPad X1 Carbon Gen 12              |

*Source:* Authors.

Table 4.5 reports the attribute values in the form in which they were collected, that is, before the CPU and Brand values are converted into the PassMark benchmark score and the Gartner market-share figure described in Section 3.4.3; that converted matrix is the input actually used

by the TOPSIS procedure of Section 4.5.

Table 4.5: Candidate laptops and their raw attribute values.

| Model                                         | Price | CPU                               | RAM | SSD  | Brand  | Weight | Battery |
|-----------------------------------------------|-------|-----------------------------------|-----|------|--------|--------|---------|
| Lenovo ThinkPad X1 Carbon Gen 13 Aura Edition | 2289  | Intel Core Ultra 7 258V           | 32  | 2048 | Lenovo | 1.09   | 57      |
| Lenovo ThinkPad X1 Carbon Gen 14 Aura Edition | 3499  | Intel Core Ultra 7 365 vPro       | 32  | 2048 | Lenovo | 1.00   | 58      |
| Apple MacBook Air M4                          | 1494  | Apple M4 (10-core)                | 24  | 512  | Apple  | 1.24   | 53.8    |
| Apple MacBook Air M5                          | 1749  | Apple M5 (10-core)                | 24  | 1024 | Apple  | 1.23   | 53.8    |
| Asus ExpertBook P5                            | 1675  | Intel Core Ultra 7 258V           | 32  | 1024 | ASUS   | 1.29   | 63      |
| HP EliteBook Ultra G1i                        | 1539  | Intel Core Ultra 7 258V           | 32  | 1024 | HP     | 1.36   | 56      |
| Lenovo ThinkPad X9 15 Aura Edition (2025)     | 1799  | Intel Core Ultra 7 258V           | 32  | 1024 | Lenovo | 1.40   | 80      |
| Dell 14 Premium (XPS 14 2026)                 | 1890  | Intel Core Ultra 7 255H           | 32  | 1024 | Dell   | 1.72   | 69.5    |
| Lenovo ThinkPad T14 Gen 6 (AMD)               | 1100  | AMD Ryzen AI 5 340                | 16  | 512  | Lenovo | 1.41   | 57      |
| Lenovo ThinkPad T14 Gen 7                     | 2357  | Intel Core Ultra 7 355 (Series 3) | 32  | 512  | Lenovo | 2.22   | 60      |
| Acer Aspire 5                                 | 395   | Intel Core i3-1115G4              | 8   | 128  | Acer   | 1.63   | 48      |
| Lenovo ThinkPad X1 Carbon Gen 12              | 1685  | Intel Core Ultra 165U vPro        | 32  | 1024 | Lenovo | 1.08   | 57      |

*Note:* price in USD, RAM and SSD in GB, weight in kg, battery capacity in Wh; attribute values as listed at the point of sale.

*Source:* Amazon.com product listings, collected 12 July 2026.

The twelve candidates span a nearly ninefold price range, from \$395 for the entry-level Acer Aspire 5 to \$3,499 for the top-specification ThinkPad X1 Carbon Gen 14 Aura Edition, and are drawn from six brands, with Lenovo contributing six of the twelve listings. Every candidate except the Acer Aspire 5 (8 GB) and the Lenovo ThinkPad T14 Gen 6 AMD (16 GB) carries 32 GB of RAM or the 24 GB unified-memory configuration used by Apple’s M-series chips, so the RAM criterion mainly separates these two candidates from the rest of the field rather than distinguishing among the other ten. The benchmark and market-share conversion applied to the CPU and Brand columns, together with the full step-by-step TOPSIS calculation, is reported in Chapter E.

## 4.5 TOPSIS Ranking Results

Applying the seven-step TOPSIS procedure of Section 3.4.4 to the converted decision matrix of Section E.1, using the AHP weights of Table 4.3, produces the closeness coefficients and ranking reported in Table 4.6. The full normalisation, weighting, and ideal-solution calculations underlying this table are reported step by step in Section E.2.

Table 4.6: TOPSIS ranking of the candidate laptops.

| Rank | Candidate                                            | $C_i$  |
|------|------------------------------------------------------|--------|
| 1    | Can9 — Lenovo ThinkPad T14 Gen 6 (AMD)               | 0.6152 |
| 2    | Can8 — Dell 14 Premium (XPS 14 2026)                 | 0.6046 |
| 3    | Can4 — Apple MacBook Air M5                          | 0.6039 |
| 4    | Can6 — HP EliteBook Ultra G1i                        | 0.5901 |
| 5    | Can3 — Apple MacBook Air M4                          | 0.5868 |
| 6    | Can5 — Asus ExpertBook P5                            | 0.5621 |
| 7    | Can11 — Acer Aspire 5                                | 0.5608 |
| 8    | Can12 — Lenovo ThinkPad X1 Carbon Gen 12             | 0.5444 |
| 9    | Can7 — Lenovo ThinkPad X9 15 Aura Edition (2025)     | 0.5416 |
| 10   | Can1 — Lenovo ThinkPad X1 Carbon Gen 13 Aura Edition | 0.5170 |
| 11   | Can10 — Lenovo ThinkPad T14 Gen 7                    | 0.4140 |
| 12   | Can2 — Lenovo ThinkPad X1 Carbon Gen 14 Aura Edition | 0.3803 |

Note:  $C_i$  is the closeness coefficient; candidate identifiers as assigned in Table 4.4.

Source: Authors.

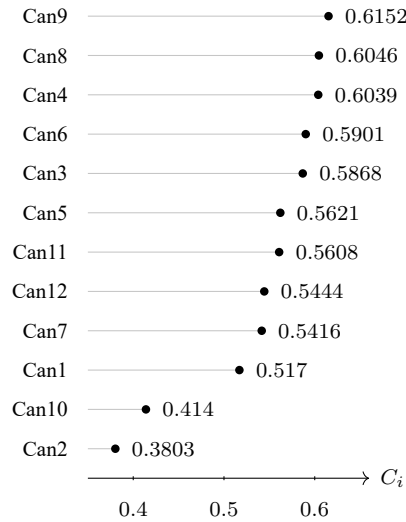


Figure 4.3: TOPSIS closeness coefficients of the twelve candidate laptops.

Source: Authors.

The Lenovo ThinkPad T14 Gen 6 (AMD) obtains the highest closeness coefficient ( $C_i = 0.6152$ ), ahead of the Dell 14 Premium ( $C_i = 0.6046$ ) and the Apple MacBook Air M5 ( $C_i = 0.6039$ ). The three leading alternatives are separated by no more than 0.011 in closeness coefficient, a narrow margin that suggests the top of the ranking may be sensitive to small changes in the underlying scores, though this was not tested through a formal sensitivity analysis. The top-ranked model reaches this position without dominating any single criterion outright. Its price of \$1,100 is the second lowest in the set, its processor score (19,546) is in the middle of the field, and its 16 GB of RAM is the smallest configuration among the eleven non-Acer candidates; it does share the joint-highest Brand score (26.5) with several other Lenovo entries in the set, but this shared strength combines with a below-median price rather than a lead on any criterion by

itself, and no single weakness is severe enough to offset that combination. The Dell 14 Premium and the MacBook Air M5 reach second and third place from a different combination, in which both carry the highest or near-highest processor scores in the field (30,684 and 26,831 respectively), which compensates for a price above the sample median.

The lowest-ranked alternative is the Lenovo ThinkPad X1 Carbon Gen 14 Aura Edition ( $C_i = 0.3803$ ), the most expensive candidate at \$3,499, despite a mid-to-upper processor score (20,946, the fourth highest in the set); because Price and CPU together account for 62.6% of the total criteria weight (Section 4.3), even a comfortably above-average score on one of these two criteria does not compensate for an extreme value on the other. The Acer Aspire 5, the cheapest candidate at \$395, ranks seventh of twelve ( $C_i = 0.5608$ ) rather than first, even though Price carries the single largest weight in the model, because its processor, memory, storage, and brand-share values are the lowest or near-lowest on every one of those criteria, and the combined deficit outweighs its price advantage. Taken together, the two extremes illustrate that no criterion, however heavily weighted, guarantees a favourable rank on its own; TOPSIS rewards a candidate's overall balance across the weighted profile rather than dominance on a single attribute. The practical implication of this result, together with its bearing on the recommended alternative, is developed further in Chapter 5.



# Chapter 5

## Discussion

### 5.1 Interpretation of the Criteria Weights

The weight structure reported in Section 4.3 concentrates 62.6% of the total priority in price (0.3491) and processor capability (0.2769), and this concentration is readable directly from the decision context the criteria were calibrated against. In the expert interviews, the purchase budget for an office machine is not a preference to be optimised but a policy set in advance by management or the IT department, so price operates as the framing variable within which every subsequent hardware trade-off is made. The dominant position of price is consistent with the wider consumer-behaviour evidence, since Liao and Chuang (2022) find price the single most influential attribute in their conjoint analysis of laptop choice, and Šostar and Ristanović (2023) identify personal budget as the strongest driver among the cultural, social, personal, and psychological factors they weigh. The weight of the processor, in turn, reflects the interviews' consistent redirection of scarce budget towards computing capability, since the experts' argument for removing the dedicated graphics card (Section 4.1.2) was precisely that the money it consumes buys more useful performance when spent on a newer-generation CPU, more memory, or faster storage.

The more interesting feature of the weight vector, however, is its bottom tier, because this is where the local calibration departs most visibly from the international literature. Battery life records the highest mean in the screening survey (4.02, Section 4.2) yet receives the joint-smallest AHP weight (0.0258, Section 4.3), and Weng Siew Lam et al. (2023) report battery life as the most heavily weighted criterion in their integrated AHP–TOPSIS model. The apparent contradiction dissolves once the two instruments are read as measuring different things. The Likert screening asks whether a criterion matters at all, and battery life clearly does, because a machine that cannot hold a charge through a meeting is unusable. The pairwise comparison instead asks how much of a decisive difference the criterion makes between candidate machines, and here the expert judgement recorded in the interviews applies. Mains power is readily available in Vietnamese offices, a runtime of four to seven hours already covers meetings and short

mobile work, and most current office laptops meet that bar. Battery life therefore behaves as a threshold requirement, broadly satisfied across the candidate set, rather than as a differentiator, and a rational pairwise comparison assigns differentiating weight elsewhere. The same logic extends to physical size and weight, which the survey ranks joint-second in importance (3.96) but the AHP places in the same low tier (0.0258). The interviews emphasised weight because motorbike commuting makes a heavy machine a daily burden, yet the office segment has converged on 14-inch machines weighing roughly 1.3 to 1.6 kilograms, so the attribute varies too little across realistic candidates to move a pairwise judgement. What the survey measures as salience, the AHP re-expresses as marginal decision value, and criteria whose requirements the market already meets lose weight in the second instrument without losing importance in the first.

A related divergence between the expert panel and the survey respondents concerns the criteria that the screening survey removed. Software compatibility was added to the candidate set on the strength of the interviews, in which office workers described a Windows lock-in effect created by accounting packages, internal ERP systems, and the government's customs declaration software (Section 4.1.2), yet the wider respondent pool screened it out, together with durability and after-sales support (Section 4.2). The likely mechanism is that the survey respondents experience compatibility as a background condition rather than a choice variable, since almost every machine they encounter in a Vietnamese office already runs Windows, so the criterion only becomes visible at the moment a non-Windows device enters consideration, which is exactly the procurement-risk scenario the experts are paid to anticipate. The two-step design of Stage 1 makes this divergence observable instead of averaging it away, and the practical reading is that the screened-out criteria function as eligibility conditions applied before the weighted comparison, not as attributes the weighted comparison should ignore.

One divergence worth flagging concerns the granularity of the Display criterion, which the screening survey of Section 4.2 removed for falling just below the retention threshold (mean = 2.80,  $SD = 1.32$ ). Read against the wider literature, this outcome may reflect how the criterion was operationalised rather than a genuine indifference to visual quality among Vietnamese office workers. Sönmez Çakır and Pekkaya (2020) decompose monitor properties into four distinct sub-criteria, resolution, size, touch-screen support, and other usage advantages, and find that these carry markedly different weights within the group, with touch-screen support outweighing resolution. Weng Siew Lam et al. (2023) go further and place screen size and pixel density in two separate higher-level categories, physical properties and technical specification respectively, with screen size ranking second within its category while pixel density ranks last within its own; the two attributes therefore answer different buyer concerns, physical footprint versus visual output quality, rather than a single underlying preference. Maghsoudi et al. (2026) report the starkest contrast, since in their sentiment-weighted ranking of sixteen notebook features, display quality places third overall, immediately behind price and design, while size places eleventh, a gap that would be invisible had the two been folded into one composite score. Set beside these findings, the single "Display" item used in the present screening instrument bundled

resolution, sharpness, anti-glare coating, and brightness into one rating, plausibly diluting a genuinely important sub-attribute, visual quality, with a less decisive one, raw screen size, and depressing the mean below the retention threshold. This is offered as an interpretive caveat rather than a claim that Vietnamese office workers disregard display quality outright, and it points to criterion granularity, not only criterion choice, as a methodological lever meriting attention in future replications of this instrument.

## 5.2 Interpretation of the Ranking

The identity of the top-ranked alternative is itself informative about what the calibrated model rewards. The Lenovo ThinkPad T14 Gen 6 (AMD) leads the ranking (Section 4.5) without dominating any single criterion outright. Its price is the second lowest in the set, its processor sits in the middle of the field, and its 16 GB of memory is the smallest configuration among the eleven non-Acer candidates; it does share the joint-highest Brand score with several other Lenovo entries in the set, but that shared strength alone does not decide the ranking. Its closeness coefficient of 0.6152 is instead the product of a profile with no severe weakness, anchored by a below-median price and reinforced by Lenovo's market-share standing on the Brand criterion. This is precisely the machine the expert interviews described before any market data had been collected, a current-generation mid-tier processor judged sufficient for administrative workloads, with the budget saved by not chasing peak specifications preserved rather than redirected into features the office segment does not use. That the data-driven ranking independently converges on this profile is a meaningful internal consistency check between Stage 1's qualitative calibration and Stage 3's quantitative outcome.

The two extremes of the ranking reinforce the same reading from opposite directions. The Acer Aspire 5, the cheapest candidate at \$395, finishes seventh rather than first despite price carrying the single largest weight, because its deficits on processor, memory, storage, and brand share accumulate faster than its price advantage compensates. At the other end, the ThinkPad X1 Carbon Gen 14 Aura Edition finishes last at \$3,499 despite a mid-to-upper processor score that ranks only fourth in the set, an outcome that echoes the over-specification argument of Rau and Fang (2018), in which paying a premium for capability beyond the requirements of the use case destroys rather than creates value once price enters the comparison with its full weight. Between these extremes, the top three alternatives are separated by less than 0.011 in closeness coefficient, so the ranking is better read as producing a short list of near-equivalent recommendations, the T14 Gen 6, the Dell 14 Premium, and the MacBook Air M5, than as declaring a single decisive winner; the Dell and the Apple machine reach their positions through the complementary route of field-leading processor scores that offset above-median prices. Within that short list, the practical tie-breakers are exactly the eligibility conditions identified in Section 5.1. The MacBook's macOS platform, for instance, would be filtered out by the Windows lock-in constraint in offices where that constraint binds, even though the weighted comparison itself

ranks it third.

### **5.3 Practical and Managerial Implications**

For an individual office worker buying with personal funds, the model's main practical export is not the specific ranking, which is tied to a catalogue snapshot, but the structure of the weights. Roughly 63% of the decision is a price–processor trade-off, storage and memory form a middle tier worth attending to, and battery life, weight, and brand deserve verification against a threshold rather than optimisation. Concretely, the buyer should fix the budget first, secure a current-generation mid-tier processor and 16 GB of memory within it, confirm that battery life and weight fall in the acceptable band for their commuting pattern, and resist paying premiums for capability at the top of the range, which the ranking shows to be counterproductive under office-work weights.

For corporate procurement, the value of the model lies in making an implicit decision auditable. The interviews indicated that purchase budgets are set by policy and that IT managers already think in terms of total cost of ownership; the AHP–TOPSIS pipeline turns those practices into an explicit weight vector and a reproducible ranking that can be attached to a purchase justification, revisited when the catalogue changes, and recalculated by replacing a single input matrix. Because the weights and the alternatives are decoupled, a procurement office can hold the weight vector stable across purchasing rounds while refreshing the decision matrix, which converts a one-off analysis into a reusable procedure. The screened-out criteria retain a role here as eligibility gates, since software compatibility and after-sales coverage are best enforced as pass-or-fail conditions before the weighted comparison, mirroring how the experts described actual procurement practice.

The results also carry signals for the supply side. For retailers advising office customers, the ranking suggests that the persuasive assortment is not the cheapest machine or the most powerful one but the balanced mid-tier configuration, and that advice framed as “sufficient processor within budget” matches how this segment actually weighs the purchase. For manufacturers, the weight structure implies that marginal engineering investment aimed at this segment earns more return in price positioning and processor-generation currency than in pushing battery life or thinness beyond the threshold the segment already considers satisfied; a battery lasting beyond the four-to-seven-hour working band, in particular, adds cost against a criterion holding under 3% of the decision weight.

### **5.4 Comparison with Prior Studies**

Set against the prior studies reviewed in Section 2.5, the present results agree on the centre of the priority structure and diverge at its edges, and the divergences are traceable to context rather than

method. The dominance of price is the most stable point of agreement, matching the conjoint evidence of Liao and Chuang (2022), the budget-driven consumer behaviour reported by Šostar and Ristanović (2023), and the notebook analysis of Maghsoudi et al. (2026), in which price, display quality, processor capability, and memory all appear near the top of the feature-priority structure. Table 5.1 summarises where the present priority structure agrees with and departs from this earlier evidence.

Table 5.1: The present priority structure set against prior studies.

| Criterion           | This study                 | Comparable prior evidence                                                                                                               | Relation   |
|---------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------|
| Price               | Highest weight (0.349)     | Most influential attribute in Liao & Chuang (2022); strongest driver in Šostar & Ristanović (2023); near top in Maghsoudi et al. (2026) | Agreement  |
| Processor           | Second weight (0.277)      | Near top of the feature-priority structure in Maghsoudi et al. (2026)                                                                   | Agreement  |
| Brand               | Low tier (0.043)           | Highest-weighted criterion in Sönmez Çakır & Pekkaya (2020)                                                                             | Divergence |
| Battery life        | Joint-lowest (0.026)       | Most heavily weighted criterion in Weng Siew Lam et al. (2023)                                                                          | Divergence |
| Graphics card       | Removed at interview stage | Ranked last among notebook features in Maghsoudi et al. (2026)                                                                          | Agreement  |
| Display             | Screened out (mean 2.80)   | Placed third in Maghsoudi et al. (2026); decomposed into sub-criteria by Sönmez Çakır & Pekkaya (2020)                                  | Divergence |
| Leading alternative | A Dell model ranks second  | Dell leads the brand ranking in Gaur (2023)                                                                                             | Consistent |

*Note:* “Agreement” and “Divergence” describe the position of the criterion in the priority structure relative to the cited study, not the numerical weight itself.

*Source:* Authors.

The clearest divergence concerns brand. Sönmez Çakır and Pekkaya (2020) find brand image the highest-weighted criterion among their six, whereas the present model assigns brand only 0.0430. The difference is explicable through the signalling mechanism that motivates brand weight in the first place. Brand compensates for information asymmetry when buyers cannot evaluate technical quality directly, and that asymmetry is largest for general consumers choosing alone. In the office-procurement context studied here, IT managers and standardised specifications absorb much of that uncertainty, so the signalling value of brand shrinks even while brand retains a residual role, visible in the ranking, where Lenovo’s market share contributes to the winning profile. The second divergence, the position of battery life relative to Weng Siew Lam et al. (2023), was examined in Section 5.1 and reflects both the device category, since a phone cannot assume mains power, and the office setting that can.

At the level of ranked outcomes, comparison is necessarily looser because candidate sets differ, but two observations hold. Gaur (2023), ranking laptop brands during the pandemic with a comparable AHP–TOPSIS apparatus, identify Dell as the leading alternative; the present ranking, built on model-level rather than brand-level alternatives, places a Dell machine second, suggesting that the earlier result survives translation to a finer-grained alternative space. Maghsoudi et al. (2026), weighting notebook features by mined sentiment, place the graphics card

last among sixteen features and price near the top, which independently corroborates both the experts' removal of the GPU from the candidate set and the weight the present model gives to price; their finding that display quality ranks third, against its removal by the present screening survey, is the one substantive disagreement, and Section 5.1 argues it is most plausibly an artefact of criterion granularity rather than a genuine difference in what office users value. Taken together, the comparisons support the position claimed in Section 2.6. The international literature travels well at the core of the priority structure, price and processing capability, and requires local recalibration precisely at the criteria, battery, weight, brand, and software ecosystem, whose value depends on infrastructure and working conditions.

# Chapter 6

## Conclusion

### 6.1 Summary of Findings

This study set out to answer three research questions about how an office worker in Vietnam should select a laptop, and each stage of the three-stage design answers one of them. In answer to RQ1, the combination of literature synthesis, expert interviews, and a screening survey of 84 respondents produced a set of seven evaluation criteria: price, processor capability, memory, solid-state storage, brand, physical size and weight, and battery life. The path to this set is itself part of the finding, since the interviews removed the dedicated graphics card and added software compatibility before the survey screened the candidate list against a pre-committed threshold, leaving a criteria set calibrated to local working conditions rather than imported from the international literature.

In answer to RQ2, the pairwise comparison of the seven criteria produced a consistent judgment matrix ( $CR = 0.0483$ , below the 0.10 acceptance threshold) whose priority vector is strongly concentrated at the top, where price carries a weight of 0.3491 and the processor 0.2769, together accounting for approximately 62.6% of the total priority, followed by storage (0.1592) and memory (0.1201), with brand (0.0430), size and weight (0.0258), and battery life (0.0258) forming a low-priority tier. The most instructive feature of this vector is the position of battery life, which recorded the highest mean in the screening survey yet the smallest pairwise weight, a reversal that Chapter 5 attributes to its status as a threshold requirement that most office laptops already satisfy.

In answer to RQ3, applying TOPSIS with these weights to twelve candidate laptops drawn from seven professional review outlets and priced from Amazon.com ranked the Lenovo ThinkPad T14 Gen 6 (AMD) first with a closeness coefficient of 0.6152, ahead of the Dell 14 Premium (0.6046) and the Apple MacBook Air M5 (0.6039). The recommended machine does not dominate any single criterion outright, sharing only the joint-highest Brand score with several other Lenovo entries, but it combines a below-median price, a mid-field processor, and that Brand standing into the most balanced profile in the set, while the cheapest and the most expensive

candidates finish seventh and last respectively, confirming that under the derived weights the office-laptop decision rewards balance rather than dominance on any one attribute.

## 6.2 Contributions

The contribution of the study is the twofold one stated in Section 1.3, now delivered rather than promised. First, the study constructs a criteria set for office laptops that is explicitly calibrated to Vietnamese working conditions, and documents where that calibration departs from priorities reported in the international literature. Battery life recedes from the top-weighted position it holds in prior device-selection studies once mains power can be assumed, device weight is salient for motorbike commuting yet varies too little across the modern office segment to differentiate candidates, and software compatibility emerges as an eligibility condition created by a Windows lock-in effect that international criteria frameworks rarely make explicit. Second, the study operationalises a complete AHP–TOPSIS pipeline on real market data, from elicited pairwise judgements through a consistency-checked weight vector to a ranking of twelve actual laptop models, with every computational step reported in the appendices in reproducible form. Beyond these two, the two-instrument design of Stage 1 carries a methodological point worth stating separately. By screening criteria with a wider respondent pool after eliciting them from experts, the study makes the divergence between the expert panel and the survey respondents observable, and Chapter 5 shows that this divergence is informative rather than noise.

## 6.3 Limitations

Four limitations bound the claims made above. First, the pairwise comparison matrix was elicited from a single representative expert, as described in Section 3.3; although the matrix passes the consistency check and the geometric-mean recomputation preserves the same criteria ordering, a weight vector aggregated over several independent experts would carry more evidential force, and the screening survey’s divergence between the expert panel and the survey respondents suggests the choice of judge is not neutral. Second, the screening sample of 84 respondents was recruited by convenience rather than probability sampling, so the retained criteria set reflects the office-worker population only to the extent that the sample does. Third, the market data are a snapshot, since prices and configurations were collected from a single retail platform on a recorded date (Section 4.4), and laptop prices in particular move quickly enough that the specific ranking, though not the method, has a short shelf life. Fourth, two criteria are measured through proxies, the processor through its PassMark benchmark score and brand through global market share, and while Section 3.4.3 argues these are the most defensible quantitative stand-ins available, neither captures its criterion completely; market share in particular measures prevalence rather than the service quality that motivated the brand criterion in the interviews. In addition, the stability of the ranking under perturbations of the weight vector was



not examined, which matters because the top three closeness coefficients lie within 0.011 of one another (Section 4.5); the ranking should accordingly be read as a short list rather than a strict ordering.

## 6.4 Future Work

The limitations point directly to the extensions worth pursuing. The most immediate is a group-decision version of Stage 2, eliciting pairwise matrices from several experts and aggregating them, which would both strengthen the weight vector and permit an analysis of inter-expert disagreement that the present single-judge design cannot support. A second extension is a formal sensitivity analysis of the ranking with respect to the weights, which the narrow margins at the top of the present ranking make particularly relevant. On the measurement side, fuzzy extensions of AHP and TOPSIS, of the kind reviewed in Sections 2.3 and 2.4, would let the model represent the imprecision of judgements such as brand trust directly instead of forcing them through crisp proxies. On the data side, the pipeline should be re-run as the Vietnamese retail catalogue catches up with the international models considered here, both to replace the Amazon.com price base with domestic prices that include local warranty terms and to test whether the ranking survives the change of market; repeating the collection at intervals would also show how quickly rankings decay as catalogues turn over. Finally, the interpretive caveat raised in Section 5.1 about criterion granularity deserves a direct test. A replication that decomposes the removed Display criterion into visual quality and screen size, following the decomposition used by Sönmez Çakır and Pekkaya (2020), would establish whether office users' apparent indifference to displays is genuine or an artefact of the composite item, and the same design could extend the model to user segments beyond office work, where the calibration developed here is not expected to hold.

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# Appendix A

## Theoretical Basis for the Candidate Criteria

This appendix sets out the academic and theoretical grounding for the fourteen candidate criteria that literature synthesis (Section 3.2.1) contributed to the candidate set, together with one additional criterion, software compatibility, that was introduced only after the expert interviews of Section 3.2.2. The discussion below is confined to prior research and general consumer-electronics theory; it deliberately excludes the interview-specific reasoning already reported for the Vietnamese context in Section 4.1.2, so that the two sources of justification, literature and fieldwork, remain visibly separate rather than blended into a single narrative.

### A.1 Economic Factors and the Budget Constraint

Price occupies a central position in the consumer decision network (Sönmez Çakır & Pekkaya, 2020). Rau and Fang (2018) add an economic framing in which price is treated as compensation for brand value, and device value is understood to depreciate as a function of time. The mechanism behind this depreciation is the pace of technological turnover: a device purchased at a high price can become functionally dated within a few years even without any physical fault, simply because the surrounding software and peripheral ecosystem has moved on. This dynamic explains why, in terms of relative priority, price behaves as a comparatively strong constraint for the general consumer segment (Weng Siew Lam et al., 2023). Price-sensitive buyers are willing to trade away hardware headroom they do not need, for instance accepting a mid-tier processor generation instead of the newest flagship chip, in exchange for a budget that still covers the tasks they actually perform (Maghsoudi et al., 2026). Expert or professional buyers, by contrast, tend to treat price as secondary, accepting a meaningfully higher cost in return for durability and dependable after-sales support. At the opposite end of the market, Jiménez-Parra et al. (2014) find that among buyers of remanufactured electronics a low price is the one incentive strong enough to make a buyer accept the residual risk of using previously owned technology.

## A.2 Core Hardware Capability

The processing capacity of the CPU together with the working memory supplied by RAM forms the microarchitectural basis for a computer’s execution capability. Because the CPU functions as the bottleneck that governs a device’s overall responsiveness, Kang et al. (2022) place it among the core quality attributes, one whose weight in the purchase decision outweighs that of accompanying after-sales services. Examined from a behavioural angle, Liao and Chuang (2022) document a shift in how consumers reason about processors: buyers increasingly favour multi-core architectures chosen not purely to maximise clock speed but to optimise performance per watt of energy consumed, a preference aligned with broader sustainability expectations. Alongside the processor, a large RAM capacity has stopped being an optional upgrade and has instead become a near-mandatory baseline, a shift driven by the requirement to keep several cloud-based platforms and resource-intensive applications running at once. This same shift toward modern hardware also explains why solid-state storage has displaced the traditional hard disk drive: because an SSD contains no spinning platter or other moving mechanical part, it removes the risk of data loss from physical shock while travelling and reduces power draw at the same time.

The graphics processing unit (GPU) exposes a clearer split between customer segments. Users with specialised needs, such as gaming or design work, require strong graphics performance and willingly accept the bulkier, noisier cooling system that a discrete GPU demands. The sentiment analysis of social-media posts conducted by Maghsoudi et al. (2026), however, shows that heat output and fan noise are among the strongest sources of complaint precisely within the office-worker segment. This rejection follows directly from a physical and an economic trade-off. Physically, a discrete GPU requires a larger cooling assembly and draws more power, which increases the device’s thickness and weight and shortens battery life, all of which work against the mobility that office use depends on. Economically, because office tasks such as word processing and spreadsheet work cannot make use of a discrete GPU’s capacity, fitting one becomes a case of over-specification: it raises the budget constraint discussed in Section A.1 without returning a proportional benefit (Rau & Fang, 2018). These two mechanisms together explain why a component that represents, on paper, superior technical capability is nonetheless declined by the general buyer segment in favour of integrated graphics.

## A.3 Portability and Durability

The ability to operate away from a fixed power source makes battery life one of the attributes buyers weigh most heavily when choosing a laptop. Weng Siew Lam et al. (2023) report that, among technical specifications, battery life carries the single highest weight in their model. Buyers oriented toward a more sustainable lifestyle place a comparable emphasis on efficient batteries as a way of lengthening the replacement cycle and limiting electronic waste (Liao & Chuang, 2022). Manufacturers nonetheless face a physical trade-off in delivering this: battery capacity scales

with the overall weight of the device.

On design and weight, the evidence runs against the assumption that younger consumers chase a slim form factor regardless of cost. The data reported by Weng Siew Lam et al. (2023) show practicality dominating aesthetics: their student respondents place durability, battery life, and processing power ahead of other attributes, pushing a slim or fashion-oriented design to the bottom of the priority ranking. Zakeri et al. (2023) reinforce this reading by treating durability as the property that determines how far a device's usable lifecycle can be extended. This body of evidence explains why buyers accept a somewhat heavier machine, or pay a premium for an impact-resistant aluminium-alloy shell, in preference to a thin plastic shell that is light but comparatively fragile (Liao & Chuang, 2022).

## **A.4 Interface and Interaction Experience**

Across long working sessions, the display and the keyboard are the two attributes that determine physical comfort and, by extension, how long a session can be sustained without fatigue. Maghsoudi et al. (2026) record that experienced users hold a demanding standard for typing feel, so that a keyboard lacking adequate key travel or tactile feedback draws negative feedback quickly and consistently.

For the display, panel technology matters alongside size. Liao and Chuang (2022) trace the decline of LCD panels to the arrival of LED backlighting, which can cut power consumption by as much as 60 percent, a gain that aligns with the broader move toward more sustainable consumption. Anti-glare coatings and blue-light filtering are likewise becoming a *de facto* standard, valued for reducing digital eye strain over extended use. Physical connectivity ports follow the opposite trajectory: rather than continuing to be upgraded, they are increasingly folded into a secondary, supplementary feature group (Sönmez Çakır & Pekkaya, 2020), a pattern that reflects the broader shift toward wireless connectivity. The growth of cloud storage together with a maturing ecosystem of Bluetooth peripherals has allowed bulkier physical ports, such as optical drives or wired network sockets, to be removed from many devices altogether, freeing up space for a more minimalist chassis.

## **A.5 Trust and Emotional Value**

Brand is a factor that shapes purchase intention directly, and Sönmez Çakır and Pekkaya (2020) treat it as a filter through which a buyer absorbs the physical and technical properties of a product without evaluating each one individually. Read through a behavioural-economics lens, this filtering role is a response to information asymmetry: because the general buyer lacks the technical background to assess a chip, a cooling assembly, or the durability of a motherboard directly, brand functions as a signalling mechanism that stands in for that missing expertise. Apple is

the clearest illustration of the mechanism: office buyers rarely investigate transistor counts or the microarchitecture of an M-series chip, and instead purchase on the strength of a logo that is trusted to guarantee smooth performance, display quality, and battery life. A less established brand offering comparable specifications on paper typically struggles to justify the same price, precisely because it lacks this trust filter to convert raw technical numbers into a sense of purchase safety. Buyers who are strongly attached to a given brand tend, as a consequence, to be comparatively insensitive to price movements elsewhere in the market (Rau & Fang, 2018). This trust function is not always carried by the original manufacturer’s logo: in the market for used or remanufactured devices, it is instead the reputation of the distributor or refurbisher that carries this role, countering the buyer’s default assumption that a used device is more likely to fail (Jiménez-Parra et al., 2014).

After-sales service is treated by buyers as a form of insurance against risk. Šostar and Ristanović (2023) find that a flexible warranty policy reduces the fear of an expensive component failure and, in doing so, builds a longer-lasting emotional attachment between the buyer and the product. Although after-sales support is often classified as a secondary attribute relative to hardware, Maghsoudi et al. (2026) show that a service failure can trigger a rapid wave of negative sentiment on social media, so that a single after-sales incident is capable of outweighing the goodwill built by genuine hardware improvements.

## **A.6 Software Compatibility (Added After the Expert Interviews)**

Unlike the fourteen criteria discussed above, software compatibility was not part of the literature-derived candidate set; it entered the candidate set only after the expert interviews of Section 3.2.2, for reasons specific to the Vietnamese office context that are reported in Section 4.1.2. Independently of that fieldwork, the wider consumer-electronics literature also treats platform or operating system as a measurable purchase attribute in its own right, rather than a background assumption. Guan and Yuen (2022) build a laptop-recommendation model in which operating system, coded across Linux, OS X, Windows 7, and Windows 8, is included as a standalone attribute, and derive distinct, internally consistent preference values for each option, treating platform choice as directly comparable to attributes such as CPU or RAM rather than as an incidental detail. Weng Siew Lam et al. (2023), summarising a smartphone-selection study by Chakraborty et al., similarly report operating system as the second most heavily weighted criterion overall, behind brand and ahead of user experience, functionality, affordability, and design. Taken together, these two sources provide an independent theoretical basis for treating software or platform compatibility as a legitimate candidate criterion for personal computing devices, a basis that stands apart from, and precedes, the Windows-based accounting and enterprise-software lock-in argument documented from the Vietnamese expert interviews in Section 4.1.2.



## A.7 Summary

Table A.1 lists each candidate criterion alongside a short description and the sources on which the discussion above draws.

Table A.1: Candidate criteria and their supporting literature.

| Criterion                 | Description                                                          | Sources                                                                                                                          |
|---------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Price                     | Purchase cost of the device                                          | (Jiménez-Parra et al., 2014; Maghsoudi et al., 2026; Rau & Fang, 2018; Sönmez Çakır & Pekkaya, 2020; Weng Siew Lam et al., 2023) |
| CPU                       | Processor capability                                                 | (Kang et al., 2022; Liao & Chuang, 2022; Maghsoudi et al., 2026; Rau & Fang, 2018; Weng Siew Lam et al., 2023)                   |
| RAM                       | Memory capacity                                                      | (Maghsoudi et al., 2026; Rau & Fang, 2018; Sönmez Çakır & Pekkaya, 2020; Weng Siew Lam et al., 2023)                             |
| SSD                       | Storage capacity and type                                            | (Liao & Chuang, 2022; Maghsoudi et al., 2026; Rau & Fang, 2018; Sönmez Çakır & Pekkaya, 2020; Weng Siew Lam et al., 2023)        |
| GPU                       | Graphics processing capability                                       | (Maghsoudi et al., 2026; Rau & Fang, 2018)                                                                                       |
| Display                   | Screen size and panel technology                                     | (Liao & Chuang, 2022; Maghsoudi et al., 2026; Sönmez Çakır & Pekkaya, 2020; Weng Siew Lam et al., 2023)                          |
| Keyboard                  | Typing feel and build quality                                        | (Liao & Chuang, 2022; Maghsoudi et al., 2026)                                                                                    |
| Connectivity ports        | Range of physical ports                                              | (Maghsoudi et al., 2026; Sönmez Çakır & Pekkaya, 2020)                                                                           |
| Design and shell material | Aesthetics and chassis material                                      | (Kang et al., 2022; Maghsoudi et al., 2026; Weng Siew Lam et al., 2023)                                                          |
| Size and weight           | Portability                                                          | (Maghsoudi et al., 2026; Weng Siew Lam et al., 2023; Zakeri et al., 2023)                                                        |
| Battery life              | Runtime on battery                                                   | (Kang et al., 2022; Liao & Chuang, 2022; Maghsoudi et al., 2026; Rau & Fang, 2018; Weng Siew Lam et al., 2023)                   |
| Durability                | Shock resistance and build life                                      | (Liao & Chuang, 2022; Weng Siew Lam et al., 2023; Zakeri et al., 2023)                                                           |
| Brand                     | Manufacturer reputation                                              | (Jiménez-Parra et al., 2014; Rau & Fang, 2018; Sönmez Çakır & Pekkaya, 2020; Weng Siew Lam et al., 2023)                         |
| After-sales               | Warranty and customer support                                        | (Kang et al., 2022; Maghsoudi et al., 2026; Šostar & Ristanović, 2023)                                                           |
| Software compatibility    | Operating-system and platform fit<br>(added after expert interviews) | (Guan & Yuen, 2022; Weng Siew Lam et al., 2023)                                                                                  |

Source: Authors.

# Appendix B

## Expert Interview Protocol and Summary

### B.1 Interview Protocol

The semi-structured interview protocol used with all five experts was constructed following the Interview Protocol Refinement (IPR) framework (Castillo-Montoya, 2016), as summarised in Section 3.2.2. The protocol comprised thirteen core questions organised into the five parts listed in Table B.1, preceded by an informed-consent script explaining the academic purpose of the study, the anonymisation of personal information, and the participant’s right to decline any question or withdraw at any time. Each session was expected to last twenty to thirty minutes and was recorded through written notes only, without audio recording.

Table B.1: Structure of the five-part interview protocol.

| Part                                | Content                                                                                              | Purpose                                                                                |
|-------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 0                                   | Greeting and informed consent                                                                        | Establish ethical basis and obtain verbal consent                                      |
| 1 — Warm-up (Q1–Q3)                 | Recent, concrete laptop-selection episode                                                            | Ground the respondent’s perspective in a specific case rather than an abstract opinion |
| 2 — Stimulus-based critique (Q4–Q7) | A literature finding for each criterion group is read aloud immediately before the matching question | Anchor agreement or disagreement to a stated academic claim                            |
| 3 — Vietnamese context (Q8–Q9)      | Open questions on locally specific factors                                                           | Surface criteria the literature synthesis could not have anticipated                   |
| 4 — Trade-off scenario (Q10–Q11)    | Hypothetical comparisons between conflicting criteria                                                | Generate qualitative seed data for the AHP pairwise comparisons of Section 3.3         |
| 5 — Closing (Q12–Q13)               | Member-checking and open addition                                                                    | Confirm the interviewer’s understanding and capture unprompted remarks                 |

*Source:* Authors.

Part 2 read a short stimulus card to the expert before each question, presenting an academic finding as material for critique rather than as a leading prompt: a card on hardware and performance reported that CPU and RAM were the two criteria rated most important by Maghsoudi et al. (2026), with GPU ranking tenth; a card on mobility and durability reported that Lam et al. (2023) ranked battery life as the most important technical-specification criterion for mobile devices; and a card on experience and trust reported that Sönmez Çakır and Pekkaya (2020) identified brand image as the most important criterion overall, alongside the finding of Maghsoudi et al. (2026) that fan noise and charger bulk drew frequent complaints yet had little influence on the final purchase decision. Most substantive questions carried a follow-up probe asking the respondent to recall the specific case behind an answer or to explain the reasoning process that produced it, a technique adapted from cognitive interviewing (Willis, 2005).

## B.2 Expert Profiles

Five experts were selected purposively so that each occupies a different position in the laptop procurement process, as summarised in Table B.2.

Table B.2: Profiles of the five interviewed experts.

| Code  | Role                                          | Experience | Position in procurement process                  | Date       |
|-------|-----------------------------------------------|------------|--------------------------------------------------|------------|
| CG-01 | Deputy head, international relations office   | 7 years    | End-user with direct purchasing experience       | 2026-07-05 |
| CG-02 | IT administrator                              | 3 years    | Specifies and maintains office device fleets     | 2026-07-09 |
| CG-03 | Import–export documentation officer           | 3.5 years  | End-user representing the target persona         | 2026-07-10 |
| CG-04 | Head of product development                   | 10 years   | Manager with organisational purchasing authority | 2026-07-08 |
| CG-05 | Retail sales and technical-support consultant | 3 years    | Advises laptop buyers daily at point of sale     | 2026-07-10 |

*Source:* Authors.

## B.3 Thematic Summary of Findings

Interview notes were analysed thematically, cross-referencing each recurring position against the two or more experts who independently raised it, consistent with the majority corroboration rule described in Section 3.2.2.

### B.3.1 Hardware Priorities and the GPU Question

Four of the five experts converged independently on removing the dedicated graphics card from the criteria set. CG-02 characterised a discrete GPU as over-specification for office work that

adds cost, weight, heat, and battery drain without a matching benefit, a position consistent with the tenth-place GPU ranking reported by Maghsoudi et al. (2026) and read to the experts as Card A. CG-03, CG-04, and CG-05 reached the same conclusion from their respective vantage points as a documentation officer, a product-development manager, and a retail consultant, each reporting that office customers who ask about GPU capability are rare and that the budget it would consume is better redirected to the CPU, RAM, or SSD. CG-05 qualified the point by noting that a discrete GPU remains justified only when the organisation has an occasional design or video-editing need, a caveat also raised by CG-02. CG-01 was the sole dissenting voice: in the direct question on GPU priority, CG-01 described integrated graphics as adequate for most office machines but characterised a discrete GPU as “nice to have, no harm in not having it,” and in the Part 4 trade-off scenario explicitly chose the GPU-equipped option, reasoning that no buyer can be certain in advance that a device will never be used for design or gaming work, so a more versatile machine is the safer choice. This single dissent is the reason the criteria revision rule in Section 3.2.2 requires majority rather than unanimous corroboration before a candidate criterion is changed. On RAM, all five experts converged on 16 GB as the new practical minimum, citing browser multitasking and simultaneous use of communication platforms such as Microsoft Teams and Zalo (CG-02, CG-03) as the drivers of memory pressure that an 8 GB configuration can no longer absorb.

### **B.3.2 Battery, Weight, and Design Trade-offs**

Four of the five experts (CG-02 through CG-05) explicitly attributed a lower priority on maximum battery capacity to the ready availability of mains power in Vietnamese offices, qualifying the emphasis placed on battery life by Weng Siew Lam et al. (2023), read to the experts as Card B; CG-01 reached the same ranking of priorities without citing mains-power access as the reason. CG-01, CG-03, and CG-05 stated that a runtime of roughly half a working day, sufficient to cover a meeting or a short period away from a desk, is adequate, while CG-04 set a more concrete range of four to seven hours. In its place, physical weight emerged as the more decisive mobility criterion: CG-01, CG-03, CG-04, and CG-05 all linked this preference to motorbike commuting and daily carrying, converging on a practical range of roughly 1.3 to 1.6 kg for a 14-inch machine. On design, CG-01, CG-03, and CG-05 associated a slim, neutral-coloured form factor with a professional impression in client-facing work, but CG-02 and CG-04 cautioned that this aesthetic preference is bounded by durability requirements, specifically hinge rigidity, thermal headroom, and resistance to chassis flex, so that appearance is treated as a secondary criterion once a minimum durability bar is met.

### **B.3.3 Brand, After-Sales, and Divergent Views on Trust**

Interview responses on brand and after-sales service were the least uniform of the five themes. CG-01 argued that the growing accessibility of AI-assisted comparison tools is reducing buy-

ers' reliance on brand reputation as a proxy for quality, a view echoed by CG-03's observation that end-users increasingly research specifications independently. CG-02, CG-04, and CG-05, however, each reported from a business-procurement or retail vantage point that brand and after-sales reliability remain decisive when device downtime carries an organisational cost: CG-02 emphasised that a business buyer facing two comparably priced machines will choose the more established brand unless an internal IT team can independently verify and service an unfamiliar one, while CG-04 and CG-05 both noted that buyers who cannot evaluate technical specifications in depth default to two or three familiar brands as a risk-management heuristic. CG-05 added that proximity of an authorised service centre to the workplace is frequently a deciding factor for organisational buyers, a point later adopted as the operational rationale for measuring Brand through a service-centre-count proxy in Section 3.4.3.

### **B.3.4 Locally Specific Factors**

All five experts identified factors specific to the Vietnamese working environment that the international literature reviewed in Section 2.5 does not address directly. The hot and humid climate was raised by CG-02 and CG-04 as a driver of thermal-management and build-quality requirements beyond what temperate-climate studies typically consider. Motorbike commuting was raised independently by CG-01, CG-03, CG-04, and CG-05 as the mechanism linking weight and durability to daily use rather than occasional travel. CG-01 and CG-03 both described a software lock-in effect arising from Windows-only accounting, customs-declaration, and internal enterprise-resource-planning software, which excludes non-Windows devices from consideration regardless of hardware merit; this observation directly motivated the addition of software compatibility as a replacement criterion for GPU, as reported in Section 4.1.2. CG-02 and CG-05 further noted that the residual prevalence of HDMI projectors and USB Type-A peripherals in Vietnamese offices keeps traditional ports relevant even as ultraportable designs move toward USB-C only.

### **B.3.5 Trade-off Scenario Responses**

When asked in Part 4 to name the three criteria they would prioritise if all others were kept flexible, core hardware performance (CPU, RAM, SSD, or a combination) appeared in every one of the five answers, making it the only criterion named by all experts. Brand or after-sales support was named by four of the five (CG-02 through CG-05), and durability was named separately by three (CG-02, CG-03, CG-04). Price, although established earlier in the interview as the first screening criterion applied by all five experts, recurred explicitly in only two of the five top-three answers (CG-01, CG-05), suggesting that once a price bracket has already narrowed the option set, experts treat it as a pre-filter rather than a continuing point of comparison. This pattern of convergence on hardware performance as the dominant criterion, together with the comparatively low priority assigned to battery life and to size or weight once the office context

was made explicit, foreshadows the weight concentration later confirmed quantitatively by the AHP results in Section 4.3.

# Appendix C

## Likert Survey Questionnaire

The screening questionnaire described in Section 3.2.3 was administered in Vietnamese through Google Forms; the English rendering below preserves its structure and wording. Respondents rated the importance of each of the fourteen candidate criteria when choosing a laptop for office work on a five-point Likert scale: 1 (not at all important), 2 (not important), 3 (neutral), 4 (important), and 5 (very important). The instrument contained no demographic items.

Table C.1: Items of the criteria-screening questionnaire.

| Item | Criterion rated                  | Scale |
|------|----------------------------------|-------|
| 1    | Price                            | 1–5   |
| 2    | Processing power (CPU)           | 1–5   |
| 3    | Memory capacity (RAM)            | 1–5   |
| 4    | Storage space (SSD)              | 1–5   |
| 5    | Display                          | 1–5   |
| 6    | Keyboard experience              | 1–5   |
| 7    | Design and shell material        | 1–5   |
| 8    | Size and weight (portability)    | 1–5   |
| 9    | Durability and shock resistance  | 1–5   |
| 10   | Battery life                     | 1–5   |
| 11   | Connectivity ports               | 1–5   |
| 12   | Software compatibility           | 1–5   |
| 13   | Brand                            | 1–5   |
| 14   | After-sales service and warranty | 1–5   |

*Note:* English rendering of the instrument administered in Vietnamese.

*Source:* Authors.

## Appendix D

# AHP Pairwise Comparison Matrix and Step-by-Step Calculation

Table D.1 reproduces the full  $7 \times 7$  pairwise comparison matrix elicited from the representative expert as described in Section 3.3. Entries above the diagonal are the 21 elicited judgements on the Saaty scale; entries below the diagonal follow from reciprocity.

Table D.1: Pairwise comparison matrix of the seven retained criteria.

|             | Price | CPU | RAM | SSD | Brand | Size/weight | Battery |
|-------------|-------|-----|-----|-----|-------|-------------|---------|
| Price       | 1     | 1   | 5   | 4   | 7     | 8           | 8       |
| CPU         | 1     | 1   | 3   | 2   | 7     | 9           | 8       |
| RAM         | 1/5   | 1/3 | 1   | 1/2 | 5     | 6           | 6       |
| SSD         | 1/4   | 1/2 | 2   | 1   | 5     | 7           | 6       |
| Brand       | 1/7   | 1/7 | 1/5 | 1/5 | 1     | 2           | 3       |
| Size/weight | 1/8   | 1/9 | 1/6 | 1/7 | 1/2   | 1           | 1       |
| Battery     | 1/8   | 1/8 | 1/6 | 1/6 | 1/3   | 1           | 1       |

*Note:* entries above the diagonal are the 21 elicited judgements on the Saaty scale; entries below the diagonal follow from reciprocity.

*Source:* Authors.

## D.1 Step 1 — Column Sums

Each column of the matrix  $A$  in Table D.1 is summed to obtain the normalising constant used in the following step. Table D.2 reports the seven column totals.

Table D.2: Column sums of the pairwise comparison matrix.

|            | Price  | CPU    | RAM     | SSD    | Brand   | Size/weight | Battery |
|------------|--------|--------|---------|--------|---------|-------------|---------|
| Column sum | 2.8429 | 3.2123 | 11.5333 | 8.0095 | 25.8333 | 34.0000     | 33.0000 |

*Source:* Authors.



## D.2 Step 2 — Weight Vector by Power Iteration (Eigenvector Method)

Following the justification given in Section 3.3 for using the principal eigenvector rather than the normalised column-average approximation (Ishizaka & Lusti, 2006; Saaty, 2003), the weight vector  $\mathbf{w}$  is obtained by power iteration: starting from the uniform vector  $\mathbf{w}^{(0)} = (1/7, \dots, 1/7)$ , each iteration computes  $\mathbf{w}^{(k+1)} = A\mathbf{w}^{(k)} / \|A\mathbf{w}^{(k)}\|_1$  until the vector stabilises. Table D.3 traces the first six iterations; the vector has already converged to four decimal places by the sixth iteration, which is taken as the final weight vector.

Table D.3: Convergence of the power-iteration weight vector across six iterations.

| Iteration | Price  | CPU    | RAM    | SSD    | Brand  | Size/weight | Battery |
|-----------|--------|--------|--------|--------|--------|-------------|---------|
| $k = 1$   | 0.2871 | 0.2618 | 0.1607 | 0.1837 | 0.0565 | 0.0257      | 0.0246  |
| $k = 2$   | 0.3550 | 0.2734 | 0.1208 | 0.1621 | 0.0405 | 0.0242      | 0.0240  |
| $k = 3$   | 0.3522 | 0.2774 | 0.1182 | 0.1581 | 0.0424 | 0.0258      | 0.0259  |
| $k = 4$   | 0.3488 | 0.2771 | 0.1201 | 0.1591 | 0.0431 | 0.0259      | 0.0259  |
| $k = 5$   | 0.3490 | 0.2769 | 0.1202 | 0.1593 | 0.0430 | 0.0258      | 0.0258  |
| $k = 6$   | 0.3491 | 0.2769 | 0.1201 | 0.1593 | 0.0430 | 0.0258      | 0.0258  |

Source: Authors.

The converged weight vector is therefore

$$\mathbf{w} = (0.3491, 0.2769, 0.1201, 0.1592, 0.0430, 0.0258, 0.0258),$$

matching the values reported in Table 4.3 of Section 4.3.

## D.3 Step 3 — Consistency Check

The converged weight vector is substituted back into  $A\mathbf{w}$ , and each component is divided by the corresponding weight to obtain seven estimates of the principal eigenvalue, as shown in Table D.4. Because  $A$  satisfies exact reciprocity and the power iteration has converged, all seven ratios agree, giving  $\lambda_{\max} = 7.3823$  directly rather than as an average over inconsistent estimates.

Table D.4: Consistency check:  $A\mathbf{w}$  and the ratio  $(A\mathbf{w})_i/w_i$ .

|                       | Price  | CPU    | RAM    | SSD    | Brand  | Size/weight | Battery |
|-----------------------|--------|--------|--------|--------|--------|-------------|---------|
| $(A\mathbf{w})_i$     | 2.5773 | 2.0444 | 0.8865 | 1.1756 | 0.3174 | 0.1903      | 0.1908  |
| $(A\mathbf{w})_i/w_i$ | 7.3823 | 7.3823 | 7.3823 | 7.3823 | 7.3823 | 7.3823      | 7.3823  |

Source: Authors.

With  $\lambda_{\max} = 7.3823$  and  $n = 7$ , the consistency index from Equation (3.3) is

$$CI = \frac{\lambda_{\max} - n}{n - 1} = \frac{7.3823 - 7}{6} = 0.0637.$$

Using the random index  $RI = 1.32$  tabulated in Table 3.2 for  $n = 7$ , the consistency ratio is

$$CR = \frac{CI}{RI} = \frac{0.0637}{1.32} = 0.0483,$$

which is below the acceptance threshold of 0.10 (Saaty, 2008), so the judgement matrix is accepted without re-elicitation, as reported in Section 4.3.

## D.4 Robustness Check — Geometric-Mean Weights

As a cross-check against the derivation method rather than against the judgement data itself, the same matrix  $A$  is also solved by the geometric-mean-of-rows method: for each row  $i$ ,  $\tilde{w}_i = \left(\prod_{j=1}^n a_{ij}\right)^{1/n}$ , and the seven values are then normalised to sum to one. Table D.5 compares the two methods.

Table D.5: Eigenvector weights versus geometric-mean weights.

|                       | <b>Price</b> | <b>CPU</b> | <b>RAM</b> | <b>SSD</b> | <b>Brand</b> | <b>Size/weight</b> | <b>Battery</b> |
|-----------------------|--------------|------------|------------|------------|--------------|--------------------|----------------|
| Eigenvector (adopted) | 0.3491       | 0.2769     | 0.1201     | 0.1592     | 0.0430       | 0.0258             | 0.0258         |
| Geometric mean        | 0.3366       | 0.2882     | 0.1185     | 0.1615     | 0.0429       | 0.0264             | 0.0259         |

*Source:* Authors.

The two methods rank all seven criteria identically and differ from one another by at most 0.013 in absolute weight, confirming that the priority structure reported in Section 4.3 does not depend on which of the two derivation methods is used.

# Appendix E

## Laptop Dataset, Benchmarks, and TOPSIS Step-by-Step Calculation

### E.1 Benchmark and Market-Share Conversion

Following Section 3.4.3, the CPU column of Table 4.5 is converted from a processor name to its PassMark CPU Mark score, and the Brand column is converted from a manufacturer name to its worldwide PC shipment market share for Q1 2026 as reported by Gartner (Gartner, Inc., 2026). Table E.1 and Table E.2 report the two conversion keys; Table E.3 then reports the full converted decision matrix that is submitted to TOPSIS.

Table E.1: CPU-to-benchmark conversion key.

| Processor                         | PassMark CPU Mark |
|-----------------------------------|-------------------|
| Intel Core Ultra 7 258V           | 18,898            |
| Intel Core Ultra 7 365 (vPro)     | 20,946            |
| Apple M4 (10-core)                | 23,622            |
| Apple M5 (10-core)                | 26,831            |
| Intel Core Ultra 7 255H           | 30,684            |
| AMD Ryzen AI 5 340                | 19,546            |
| Intel Core Ultra 7 355 (Series 3) | 20,452            |
| Intel Core i3-1115G4              | 5,811             |
| Intel Core Ultra 7 165U (vPro)    | 16,676            |

*Note:* one entry per distinct processor in the candidate set.

*Source:* PassMark Software (2026).

### E.2 TOPSIS Step-by-Step Calculation

Given the converted decision matrix of Table E.3 and the AHP weight vector of Table 4.3, the seven steps of Section 3.4.4 are carried out as follows.

Table E.2: Brand-to-market-share conversion key.

| Brand  | Market share (%) |
|--------|------------------|
| Lenovo | 26.5             |
| HP     | 19.3             |
| Dell   | 16.5             |
| Apple  | 10.6             |
| ASUS   | 6.7              |
| Acer   | 6.4              |

*Note:* worldwide PC shipment market share, first quarter of 2026.

*Source:* Gartner, Inc. (2026).

Table E.3: Converted decision matrix submitted to TOPSIS.

| Candidate | Price | CPU   | RAM | SSD  | Brand | Weight | Battery |
|-----------|-------|-------|-----|------|-------|--------|---------|
| Can1      | 2289  | 18898 | 32  | 2048 | 26.5  | 1.09   | 57      |
| Can2      | 3499  | 20946 | 32  | 2048 | 26.5  | 1.00   | 58      |
| Can3      | 1494  | 23622 | 24  | 512  | 10.6  | 1.24   | 53.8    |
| Can4      | 1749  | 26831 | 24  | 1024 | 10.6  | 1.23   | 53.8    |
| Can5      | 1675  | 18898 | 32  | 1024 | 6.7   | 1.29   | 63      |
| Can6      | 1539  | 18898 | 32  | 1024 | 19.3  | 1.36   | 56      |
| Can7      | 1799  | 18898 | 32  | 1024 | 26.5  | 1.40   | 80      |
| Can8      | 1890  | 30684 | 32  | 1024 | 16.5  | 1.72   | 69.5    |
| Can9      | 1100  | 19546 | 16  | 512  | 26.5  | 1.41   | 57      |
| Can10     | 2357  | 20452 | 32  | 512  | 26.5  | 2.22   | 60      |
| Can11     | 395   | 5811  | 8   | 128  | 6.4   | 1.63   | 48      |
| Can12     | 1685  | 16676 | 32  | 1024 | 26.5  | 1.08   | 57      |

*Note:* CPU as PassMark CPU Mark score (Table E.1); Brand as market share in % (Table E.2); price in USD; RAM and SSD in GB; weight in kg; battery capacity in Wh; candidate identifiers as assigned in Table 4.4.

*Source:* Authors.

**Step 2 — Vector normalisation.** Following Equation (3.4), each column of Table E.3 is divided by the Euclidean norm of that column across the twelve candidates. Table E.4 reports the seven column norms; Table E.5 reports the resulting normalised matrix  $r_{ij}$ .

| Table E.4: Euclidean norms of the criterion columns. |         |          |       |         |       |        |         |
|------------------------------------------------------|---------|----------|-------|---------|-------|--------|---------|
|                                                      | Price   | CPU      | RAM   | SSD     | Brand | Weight | Battery |
| Norm                                                 | 6672.10 | 72103.18 | 98.31 | 3934.83 | 71.89 | 4.94   | 207.70  |

Note: each norm is  $\sqrt{\sum_i x_{ij}^2}$  taken over the twelve candidates.

Source: Authors.

| Table E.5: Normalised decision matrix $r_{ij}$ . |        |        |        |        |        |        |         |
|--------------------------------------------------|--------|--------|--------|--------|--------|--------|---------|
| Candidate                                        | Price  | CPU    | RAM    | SSD    | Brand  | Weight | Battery |
| Can1                                             | 0.3431 | 0.2621 | 0.3255 | 0.5205 | 0.3686 | 0.2206 | 0.2744  |
| Can2                                             | 0.5244 | 0.2905 | 0.3255 | 0.5205 | 0.3686 | 0.2024 | 0.2792  |
| Can3                                             | 0.2239 | 0.3276 | 0.2441 | 0.1301 | 0.1474 | 0.2510 | 0.2590  |
| Can4                                             | 0.2621 | 0.3721 | 0.2441 | 0.2602 | 0.1474 | 0.2490 | 0.2590  |
| Can5                                             | 0.2510 | 0.2621 | 0.3255 | 0.2602 | 0.0932 | 0.2611 | 0.3033  |
| Can6                                             | 0.2307 | 0.2621 | 0.3255 | 0.2602 | 0.2684 | 0.2753 | 0.2696  |
| Can7                                             | 0.2696 | 0.2621 | 0.3255 | 0.2602 | 0.3686 | 0.2834 | 0.3852  |
| Can8                                             | 0.2833 | 0.4256 | 0.3255 | 0.2602 | 0.2295 | 0.3481 | 0.3346  |
| Can9                                             | 0.1649 | 0.2711 | 0.1628 | 0.1301 | 0.3686 | 0.2854 | 0.2744  |
| Can10                                            | 0.3533 | 0.2836 | 0.3255 | 0.1301 | 0.3686 | 0.4493 | 0.2889  |
| Can11                                            | 0.0592 | 0.0806 | 0.0814 | 0.0325 | 0.0890 | 0.3299 | 0.2311  |
| Can12                                            | 0.2525 | 0.2313 | 0.3255 | 0.2602 | 0.3686 | 0.2186 | 0.2744  |

Source: Authors.

**Step 3 — Weighted normalised matrix.** Following Equation (3.5), each column of Table E.5 is multiplied by the corresponding AHP weight from Table 4.3, giving the weighted matrix  $v_{ij}$  in Table E.6.

**Step 4 — Ideal solutions.** Following Equation (3.6), for each of the seven columns of Table E.6, the positive ideal solution  $A^+$  takes the maximum value for a benefit criterion (CPU, RAM, SSD, Brand, Battery) and the minimum for a cost criterion (Price, Weight);  $A^-$  takes the opposite extreme in each case. Table E.7 reports both vectors.

**Steps 5–7 — Separation measures, closeness coefficient, and ranking.** Following Equation (3.7), for each candidate,  $S_i^+$  and  $S_i^-$  are the Euclidean distance from its row of Table E.6 to  $A^+$  and  $A^-$  respectively. Table E.8 reports both distances for all twelve candidates. Substituting  $S_i^+$  and  $S_i^-$  into Equation (3.8) gives the closeness coefficient  $C_i$  for each candidate; the resulting values and final ranking are reported in Table 4.6 of Section 4.5.

Table E.6: Weighted normalised decision matrix  $v_{ij}$ .

| Candidate | Price   | CPU     | RAM     | SSD     | Brand   | Weight  | Battery |
|-----------|---------|---------|---------|---------|---------|---------|---------|
| Can1      | 0.11978 | 0.07258 | 0.03910 | 0.08287 | 0.01585 | 0.00569 | 0.00708 |
| Can2      | 0.18309 | 0.08045 | 0.03910 | 0.08287 | 0.01585 | 0.00522 | 0.00721 |
| Can3      | 0.07818 | 0.09073 | 0.02932 | 0.02072 | 0.00634 | 0.00648 | 0.00668 |
| Can4      | 0.09152 | 0.10305 | 0.02932 | 0.04143 | 0.00634 | 0.00642 | 0.00668 |
| Can5      | 0.08765 | 0.07258 | 0.03910 | 0.04143 | 0.00401 | 0.00674 | 0.00783 |
| Can6      | 0.08053 | 0.07258 | 0.03910 | 0.04143 | 0.01154 | 0.00710 | 0.00696 |
| Can7      | 0.09414 | 0.07258 | 0.03910 | 0.04143 | 0.01585 | 0.00731 | 0.00994 |
| Can8      | 0.09890 | 0.11785 | 0.03910 | 0.04143 | 0.00987 | 0.00898 | 0.00863 |
| Can9      | 0.05756 | 0.07507 | 0.01955 | 0.02072 | 0.01585 | 0.00736 | 0.00708 |
| Can10     | 0.12334 | 0.07855 | 0.03910 | 0.02072 | 0.01585 | 0.01159 | 0.00745 |
| Can11     | 0.02067 | 0.02232 | 0.00977 | 0.00518 | 0.00383 | 0.00851 | 0.00596 |
| Can12     | 0.08817 | 0.06405 | 0.03910 | 0.04143 | 0.01585 | 0.00564 | 0.00708 |

Source: Authors.

Table E.7: Positive ( $A^+$ ) and negative ( $A^-$ ) ideal solutions.

|       | Price   | CPU     | RAM     | SSD     | Brand   | Weight  | Battery |
|-------|---------|---------|---------|---------|---------|---------|---------|
| $A^+$ | 0.02067 | 0.11785 | 0.03910 | 0.08287 | 0.01585 | 0.00522 | 0.00994 |
| $A^-$ | 0.18309 | 0.02232 | 0.00977 | 0.00518 | 0.00383 | 0.01159 | 0.00596 |

Source: Authors.

Table E.8: Separation measures  $S_i^+$  and  $S_i^-$  for each candidate.

| Candidate | $S_i^+$ | $S_i^-$ |
|-----------|---------|---------|
| Can1      | 0.10900 | 0.11667 |
| Can2      | 0.16670 | 0.10228 |
| Can3      | 0.09002 | 0.12784 |
| Can4      | 0.08458 | 0.12897 |
| Can5      | 0.09165 | 0.11763 |
| Can6      | 0.08591 | 0.12369 |
| Can7      | 0.09575 | 0.11311 |
| Can8      | 0.08882 | 0.13579 |
| Can9      | 0.08631 | 0.13799 |
| Can10     | 0.12647 | 0.08934 |
| Can11     | 0.12725 | 0.16245 |
| Can12     | 0.09579 | 0.11449 |

Note: candidate identifiers as assigned in Table 4.4.

Source: Authors.